Collaborator or Backup: Exploring the Role of AI Notetakers in Physical Classroom Settings

An HCI study about perceptions of AI Notetakers by University Students in classrooms

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This thesis explores university students' perceptions of AI-Notetakers in physical classroom settings, an area that remains underexplored despite the tools' adoption in professional online contexts. Grounded in the field of Human-Computer Interaction (HCI), this study investigates how these systems are situated within in-person academic environments and their potential to support students during lectures and collaborative activities. Using a qualitative approach, the research explores the expectations, needs, and ethical concerns of students across technological disciplines. The findings reveal that student responses vary by linguistic background, positionality, and prior experience. Participants generally perceived AI Notetakers as collaborative partners and post-lecture sensemaking tools rather than replacements for active learning. They envisioned significant value in features like mind-mapping and dashboard views to support comprehension in challenging learning situations. However, the study also uncovers critical concerns regarding the ethical integration of AI in shared spaces. Students highlighted risks associated with hallucination, the disruption of established classroom norms, and the vital importance of institutional and peer consent. The thesis concludes by proposing design implications focused on preserving student agency, enabling customization, and ensuring epistemic transparency. Ultimately, this research emphasized the need for ethically acceptable frameworks to guide the use of AI Notetaking tools in physical higher education environments.

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CCS CONCEPTS • Human-Computer Interaction • Human-AI Interaction • Educational AI

Additional Keywords and Phrases: Note-Taking, AI Notetakers, Student Agency, Educational AI, Automatic Speech Recognition, Cognitive Load, AI Tools in Classrooms, LLM.

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1 INTRODUCTION

For many university students, the classroom is the site of a persistent cognitive tug-of-war where every second spent capturing information is a second stolen from actually understanding it. This struggle is universal, but it is significantly intensified for students navigating these pressures in a non-native language, where every missed word represents a potential gap in understanding [21, 30, 48]. Note-taking remains a cornerstone of academic success, influencing students' behavior and learning. Research consistently indicates that up to 99% of university students engage in note-taking practices during lectures, with 94% identifying it as an essential component of their educational experience [4]. Despite its importance, students are often unskilled at creating coherent notes rarely formally taught, leaving students to develop individual strategies to cope with cognitive and linguistic pressures [30]. From a cognitive perspective, this simultaneous act of listening, processing, and recording imposes substantial mental demands on learners [36, 45].

Traditionally, students have relied on different modalities to bridge these gaps [6, 22]. However, the recent emergence of AI Notetakers introduces a profound shift in the contemporary classroom and raises important questions. If note-taking has traditionally been understood as an active cognitive process that supports learning [36], what happens when AI systems begin to capture and structure lecture content automatically? How do students interpret the presence of AI Notetakers in shared physical classroom spaces? Do they view such tools as supportive, intrusive, empowering, or potentially disruptive?

Existing research provides valuable but fragmented insights examining cognitive processes involved in note-taking [36, 45], input modalities [6, 22], and AI tools in educational or professional contexts [19, 41]. Critical scholars have also questioned the broader implications of AI integration in education by involving students and teachers [7, 8, 42]. However, limited empirical attention has been paid to how students would perceive these AI actors, termed as AI Notetakers, specifically within in-person classroom settings, where social norms, recording policies, and peer dynamics are directly implicated. From a Human–Computer Interaction (HCI) perspective, emerging technologies must be examined not only in terms of their technical capabilities but also in how users make sense of, negotiate, and integrate them into everyday practices [4, 36, 37]. The success or failure of AI note-takers in classrooms will depend not solely on accuracy or efficiency, but on how students make sense of their role within learning processes and classroom ecosystems.

Aim and Research Question: The aim of this thesis is to explore how students in university make sense of these new technological actors within physical classroom environments. To achieve this, the study addresses a primary Research Question:

How do university students perceive the presence and role of AI Notetakers in physical classroom settings?

This is explored through two sub-questions:

- SQ1: What are the expectations and potential benefits students associate with such tools?
- SQ2: What are the concerns and challenges they identify regarding their responsible and acceptable use?

Drawing on **Weick's Sensemaking Theory** [49] as the primary analytical lens, this study examines how students construct meaning around a novel and unfamiliar technology in a shared learning space. By foregrounding students' perspectives, this study contributes to ongoing discussions about AI in education and offers insight into how AI-mediated note-taking may reshape learning practices in higher education.

SIGNIFICANCE: This work makes a theoretical contribution to the fields of Human-Computer Interaction and Human-AI interaction, deepening understanding of students' perspectives on AI Notetakers in physical classroom settings. While AI tools in education have received significant attention, research specifically examining students' perceptions of AI Notetakers in the classroom environment remains limited. This study argues that university students envisioned conditionally accepting AI Notetakers not as substitutes for their engagement but as identity collaborators and learning scaffolds — a framing rooted in participants' own relational metaphors such as 'assistant,' 'friend,' and 'collaborator.' The potential acceptability depends on the extent to which they would preserve student agency, support active learning, and respect the social norms of shared classroom spaces. Through a sensemaking lens, this study suggests that, regardless of individual note-taking practices, students understood their active presence in the learning process as a core part of their academic identity and their perceptual responses are best understood as attempts to protect that presence rather than simply evaluate a new tool. Students' envisionment varied considerably, shaped by their prior educational experiences, linguistic backgrounds, and individual positionality. The findings offer practical insight for HCI designers, educators, institutional policymakers, and researchers navigating the integration of AI-mediated tools into shared pedagogical ecosystems.

Delimitation: This thesis is delimited to exploring the perceptions of university students regarding AI Notetakers within physical classroom settings. The study focuses on students from technological backgrounds, given their familiarity with AI tools, and is conducted with non-native English speakers. It does not involve the design, implementation, or technical performance evaluation of AI notetaking systems.

Thesis Outline: The rest of the paper is organized as follows. Section 2 provides background research on notetaking, classroom challenges, AI tools in education, and the emergence of AI Notetakers. Section 3 presents the research methodology and the data collection method, and Section 4 discusses the Results and Analysis. Section 5 presents the discussion drawing on the findings that answer the main RQ and sub-questions, as well as the contributions of this research, and discusses the limitations and design implications. Finally, section 6 concludes this paper, followed by section 7, the acknowledgment.

2 BACKGROUND AND THEORY

2.1 Note-taking and Cognitive Load

Note-taking is widely recognized as a fundamental academic activity that most students engage in, as it supports information tracking, concentration, learning, and enhancing exam performance [6, 25]. In higher education, where lectures comprise nearly 83% of teaching, students habitually take notes in class, even without explicit instruction [6, 36]. Piolat compared the cognitive complexity of note-taking to playing chess, as both require simultaneous knowledge retrieval, careful planning, and problem-solving [25, 36]. Beyond its cognitive demands, note-taking is also a material and embodied practice where the body, the pen, and the page participate together in the construction of meaning [6, 36].

This process involves two mechanisms: *encoding* (active processing during capture) and *external storage* (review). However, effective note-taking is not merely transcription; it requires prioritizing crucial concepts while excluding extraneous details [4]. When learners take notes, they are actively processing, synthesizing, and summarizing core concepts in their own words, which requires a deeper level of comprehension [4], creating a substantial cognitive load [6].

Despite its ubiquity, note-taking is not an innate skill, nor is it systematically taught in many educational institutions. Nevertheless, good note-taking practices are associated with more effective study strategies and improved academic performance [4, 14, 30].

Prior research has shown that students customize their notes in their own preferred ways, leading to diversification in note-taking. For instance, some students create their own abbreviations or use commonly accepted ones, highlight significant points with different colors, or, if typing, use varied font colors and styles. These individual adaptations arise in response to the cognitively demanding nature of tasks, which involve capturing information from one or more sources simultaneously [8, 36]. In classrooms, students are the primary actors who identify the context, translate spoken words into concise, meaningful, and often abbreviated notes[6].

Environmental and contextual factors also influence this load. Research confirms that notetaking is context-dependent, meaning that the physical environment, peer distractions, and multitasking behaviors significantly impact the quality of the notes produced, and depth of the learning that accompanies them [14, 36, 45]. Linguistic proficiency further shapes the cognitive effort required [36]. To date, lecture halls and classrooms remain the most common places where students are immersed in the act of learning and note-taking [14].

Together, these cognitive, environmental, and linguistic pressures make the physical classroom a particularly demanding context for note-taking, in which technological additions have crucial implications for how students learn and engage.

2.2 Evolution from Traditional to Digital Notes and Different Modalities

Technological advancements have transformed note-taking modalities; though handwritten and digitally typed notes remain the most prevalent [30]. The increasing integration of technology into education has led to the use of digital devices like laptops and tablets in classrooms and their incorporation into students' learning experiences [6, 22].

Comparative studies by Friedman highlight a tension between volume and depth; while typing facilitates greater content capture with reduced cognitive load, it often encourages passive, verbatim transcription notes produced positively [6, 14, 30]. In contrast, handwriting's slower pace forces selective note-taking, promotes summarization, and deeper cognitive encoding [4, 14, 30]. Digital methods offer additional benefits such as enhanced searchability, ease of editing and sharing, and greater flexibility [16, 22], leading to hybrid note-taking systems that attempt to merge the benefits of both laptop and pen-and-paper methods, though these require further investigation [14].

Beyond traditional input modalities like keyboard, digital ink, and voice afford distinct interactions that potentially influence learners' note-taking behavior and learning outcomes. Voice input is emerging as a viable alternative, driven by advances in Automatic Speech Recognition(ASR) systems in the market [22, 47]. Speaking requires less cognitive effort than writing, providing a more immediate channel for documenting fleeting ideas before they are lost [22].Ultimately, the emergence of ASR lays the technical groundwork for **AI Notetakers** —a new category of tools designed to capture and structure content entirely.

2.3 Classroom Challenges and Non-native English Speakers

The cognitive demands of note-taking are compounded in the university classroom, where linguistic diversity, lecture pace, and limited input control create additional barriers for students. Language is often the most important aspect when delivering lectures or taking notes. English is the predominant language in numerous domains, including academia. This dominance poses difficulties for non-native English speakers. Lower proficiency in English may affect conciseness and fluency, often leading to grammatical inaccuracies. Limited linguistic proficiency can increase cognitive load, making it harder for students to listen, understand, and record information in real time [21]. Effective note-taking depends not only on a student's ability to listen and record information but also on the lecturer's skill in delivering the material [30].

Friedman, citing Ewer, identifies three core barriers to effective note-taking: *language and speech* (difficulties with speed and accent), *concepts* (mental load of capturing novel ideas), and *mental set* (habits of oversimplification)[14]. These barriers are particularly acute for Non-Native English speakers **(**NNEs), for whom language adds an extra layer of cognitive difficulty. Studies on NNEs report that students struggle to follow lectures delivered at high speed, to recognize unfamiliar vocabulary and discourse markers, and to write notes while simultaneously processing spoken input in a second language [21]. Malavska notes that students studying in their native language face significantly fewer challenges when taking notes compared to those studying in English. Consequently, citing Dunkel, the study emphasizes that mastering simultaneous listening and note-taking is a matter of "academic survival" for NNE students in English-medium universities [30].

Listening differs fundamentally from reading due to the lack of control over input pace; students cannot pause live lectures, leaving comprehension highly vulnerable to a speaker's pronunciation, dialect, and speech clarity. Trzeciak and Mackay, alongside Al-Musalli, distinguish between these modalities by noting that while reading affords individuals greater control to pause and review text, listening is a continuous, non-stop process in which the listener cannot halt the speaker to rehear the context. Furthermore, students' real-time experience is influenced, positively or negatively, by the speaker's pauses, false starts, or unclear pronunciation [2, 14].

Evidently, to overcome these constraints, NNE students frequently record key terms in English but switch to their native language, citing accent and speech as major barriers to comprehension [30]. These classroom realities illustrate that note-taking challenges stemming from both cognitive complexity and linguistic barriers create immense possibilities for technological interventions. To cope with these challenges, NNEs frequently rely on technological support such as machine translation and AI writing assistants [21, 48], which helps them formulate ideas and write in English. Recent research conducted on an AI rewriting tool suggested an existing openness to AI support, a context directly relevant to understanding how they might perceive AI notetakers in physical classrooms [21].

2.4 AI in education and classroom AI tools (AIEd)

Artificial Intelligence (AI) presents significant opportunities for tackling major challenges in education and society [8]. The education sector is currently experiencing what Selwyn describes as *"AI fever,"* marked by a significant increase in students' use of ChatGPT and other generative AI tools [8, 42, 48]. AI is now integrated into institutional management, teaching support, and student assistance, encompassing automated grading, online exam proctoring, teacher-student interaction chatbots, and classroom surveillance for student assessment [42], Natural Language Processing (NLP) for writing assistance, and personalized learning systems that adapt materials based on past performance[21, 42, 48].

Large Language Models (LLMs), built on transformer architectures, have rapidly emerged as indispensable and are widely adopted due to their capacity to understand, process, and generate human-like text with remarkable fluency and coherence [19]. These capabilities are particularly relevant in educational contexts, where tasks such as summarization, comprehension support, and content structuring align closely with LLMs' core strengths. Conversely, a known limitation is hallucination – the tendency of AI systems to generate inaccurate or fabricated content with unwarranted confidence, leading to a loss of trust in AI tools [10, 42].

Schools and AI technologies share a structural similarity, both relying on processes of monitoring, categorizing, standardizing, synchronizing, and sorting. Within Human-Computer Interaction (HCI), the classroom ecology represents the interconnected

system of social interactions, physical space, and tools where the introduction of a new device inevitably reshapes behaviors, power dynamics, and learning norms of the entire environment [29]. Approaching this delicate ecology primarily through the lens of data capture raises concerns, as it suggests several fundamental shifts in educational norms [42]. Despite these tensions, past studies have examined the potential of classroom chatbots to improve accessibility and social inclusion for international students[8]. It is within this broader context that AI-assisted note-taking tools have begun to emerge as a distinct application of LLM capabilities in academic environment.

2.5 AI-Based Note-Taking: Paving Path for AI Notetakers

Building upon advances in ASR and LLMs, AI-assisted note-taking tools have emerged as a distinct category. Earlier systems, such as Stupad, linked handwritten notes to audio streams, reducing verbatim transcription and encouraging summarization [4, 22]. More recent tools integrate automatic speech recognition and summarization to produce real-time transcripts and structured notes [22, 23, 47].

These parallel advancements in LLM technology have accelerated the development of sophisticated, automated note-taking tools[19, 48]. Previous research examined AI Notetakers in professional settings, focusing on their ability to record conversations and capture important points[41].

Within this broader landscape, AI-assisted note-taking tools aim to reduce students' cognitive load by automatically capturing and structuring lecture content. Tools such as NoteGPT [32], Otter [33], and Notability [22] exemplify the practical realization of LLM, improving the efficiency and effectiveness of extracting key information. These tools are designed to move students from passive learning to active, self-directed learning by enabling recording, transcription, and summarization of lectures [19].

However, few studies have explored how LLM-based systems, referred to in this study as AI Notetakers, can be used alongside traditional handwritten and typed notes in physical classroom settings. While AI notetakers may reduce cognitive and linguistic burdens, they also raise questions regarding privacy, responsible use, and impact on active engagement. As this domain remains understudied, it is essential to address this research gap by investigating how university students perceive the presence of AI Notetakers in the classroom environment.

2.6 Theoretical Framework

This study applies Karl Weick's **Sensemaking Theory** [26, 49], to explore how university students interpret the role of AI Notetakers in physical classrooms. Sensemaking is how individuals derive meaning from ambiguous or novel experiences [43, 49], operating through two complementary lenses: a *cognitive view* of information processing, and a *sociocultural orientation* where meaning-making is a collective process [26, 27, 49].

Central to this study are Weick's seven properties of sensemaking [49], which provides the analytical structure [35] for the data. As AI Notetakers are a novel technology in the classroom, these properties provide a lens for understanding how students envision their potential role:

1. Identity Construction: How students redefine and envision their role as active or passive learner.
2. Retrospective: Making sense of the present by looking at past events (e.g., evaluating AI Notetaker based on past experience using AI tools) and previous note-taking struggles to ground their expectations of the new tool.
3. Enactment: How students imagined actions such as their intent to stop taking manual notes or keep a check over AI tool shape the perceived social reality of the classroom.
4. Social: The social nature of meaning-making implying how perceptions of AI are influenced by peer reactions and lecturer policies.
5. Ongoing: The continuous evolution of perceptions over time like anticipated shift in trust as they use the tool.
6. Extracted cues: Small signals students notice in their mental image of the tool to build a larger story. (e.g. using visual cues for learning, or recording icon)
7. Plausibility: Prioritizing a trustworthy account such as accepting gaps over fabricated accuracy.

This lens aligns with intellectual agency[28, 51] —the student's ability to maintain control and critical oversight over their thinking and decision-making process while interacting with technology. Collaboration, in this context, is understood not in

its formal sense of joint task completion, but as a perceived relational metaphor in which student agency is maintained alongside the delegation of specific cognitive functions to the tool [28, 51]. As Poquet [37] suggests, sensemaking is ideal for examining experiences in "unknown" or "intuitive" situations where technology is underexplored. Following previous HCI research that applied sensemaking to Learning Management Systems [27], Generative AI [44], ChatGPT usage in classrooms [48], this study focuses on participants' narratives and framings of AI notetakers, rather than testing a specific prototype. This approach allows for a deep exploration of student interpretations to inform the design of future classroom AI support tools.

3 METHODOLOGY AND METHOD

3.1 Research Design and Rationale

This study adopts a qualitative, interpretivist research design to explore university students' perceptions of AI Notetakers in physical classroom settings [18]. Exploratory approaches are particularly appropriate for investigating emerging technologies, where social norms are still forming [39]. Utilizing scenario-based and experience-centered methods elicits reflections on technologies that are not yet fully integrated into everyday contexts [3]. Sensemaking Theory [49] provides analytical lens to capture the mental models students construct when envisioning AI notetakers in their academic environments.

3.2 Participants and Sampling Strategy

The recruitment process was guided by the Design Cycle of the qualitative research framework established by Hennink et al.[18]. This study employed a purposive sampling strategy to identify 'information-rich' participants who could provide deep insights into the sociotechnical nuances of AI integration [18, 34]. Inclusion criteria required participants to have experience attending in-person lectures, utilize English for note-taking, and be familiar with generative AI tools, ensuring they could meaningfully ground their reflections in existing academic practices. Recruitment was facilitated through university mailing lists, student communities, and advertisements. A total of 14 university students (N=14) were recruited, predominantly from technological disciplines (e.g., HCI, Machine Learning, Computer Science), with two participants from social sciences (Psychology and Gender Studies), who held prior technical backgrounds (Graphic Design and Computer Science), ensuring AI literacy across the cohort.

All 14 participants were non-native English speakers (NNEs), their functional relationship with English varied based on their prior education. To capture a spectrum of linguistic experiences, the sample was further categorized by the participants' prior **Medium of Instruction (MoI)**:

- **English as Secondary MoI** (n=10): Previous education primarily in their native languages, navigating English as a secondary medium for their current Master's studies.
- **English as Primary MoI** (n=4)**:** Non-native speakers, who had spent a significant portion of their formative education (10+ years) or entire university years in English medium of instruction.

3.3 DATA COLLECTION

3.3.1 Semi-Structured Interviews:

Primary data consisted of 14 semi-structured interviews (45 to 60 minutes), conducted via Zoom (n=9) and in person (n=5), based on participants' preferences, ensuring a naturalistic setting for conversation. This method was particularly appropriate as it allowed for deep articulation of reasoning processes and contextual experiences [9, 18].

3.3.2 The Interview Cycle and Thematic Blocks:

The interview guide transitioned from rapport building to critical reflection [18]. Its design was informed by literature regarding cognitive language barriers in second-language higher education, and classroom AI integration, and was iteratively validated through four pilot interviews. These pilots ensured that questions effectively elicited deep reflections without being leading.

Interviews began with '*contextual mapping'* of current note-taking habits, struggles, and linguistic barriers to ground the data in the students' real-life experiences. This was followed by a *'scenario-based prompt'* – a conceptual description of an AI notetaker capable of real-time transcription and summarization. This provided a shared reference point, essential for the sensemaking of a speculative technology, enabling a projective reflection on how such tools might impact classroom norms

and personal agency. The final phase utilized *'provocative social probes'* to explore anticipated behavior changes and ethical responsibilities regarding peer dialogue. The specific thematic objectives and focus areas of the interview guide are detailed in Table 1.

Table 1: Interview Structure and Thematic Blocks

Thematic Block	Focus Area	Key Objectives
Current Practices and Barriers	Baseline Habits	Mapping note-taking strategies and AI tool usage; identifying cognitive/ linguistic struggles in EMI settings.
The AI Notetaker Scenario	Speculative Utility	Eliciting reactions to transcription/summarization; exploring anticipated benefits and shifts in student agency.
Classroom Ecology and Responsibility	Social Dimensions	Investigating privacy, trust, and hesitation factors; uncovering shifting norms regarding recording and peer dialogue.

3.4 DATA PREPARATION AND PROCESSING

Transcription Workflow: Interviews were transcribed using Gladia (GDPR-compliant), EU-based data processing, and high accuracy for non-native English accents [15]. To preserve complete data sovereignty, all the audio files were downloaded to a password-protected device, and permanently removed from the platform.

Verification and Intelligent Verbatim Transcription: Manual verification for speaker diarization errors and capture nuances of second-language speech. To improve readability, the study used an intelligent verbatim [20] technique that eliminates fillers (like "um," "uh") and repeated words from transcripts. To verify linguistic accuracy, the researcher simultaneously reviewed the text and listened to the original audio; technical and accented terms received particular attention during verification. This human-in-the-loop process ensured that the distinctive narratives of non-native speakers were preserved while maintaining an anonymized audit trail (P1-P14).

Anonymization: Data were pseudonymized, with participants assigned alphanumeric codes (P1–P14). Identifiers, such as peer names or specific courses, were redacted to protect complete anonymity.

Familiarization and Memo Creation: Data preparation was treated as the fundamental phase of the analysis. Frequent listening to recordings and reviewing the transcripts allowed for deep immersion in the data and helped rectify mistakes [18]. *Analytical memos* were also utilized to record "first impressions," or surprising participant responses, and initial reflexive ideas during this process. These memos acted like a bridge between the raw data and subsequent themes, capturing the emotional tone and social context of the interviews that text alone would have missed.

3.5 DATA ANALYSIS

Data analysis followed the Thematic Analysis approach described by Braun and Clarke [5]. Unlike codebook-based approaches to thematic analysis, which rely on predetermined fixed codes applied consistently across data, this approach prioritizes the researcher's engagement with the data to maintain a unified interpretive perspective. Guided by Hennink et al [18], an abductive approach was implemented, using Sensemaking Theory [49] as an analytical lens during the coding phase, allowing the researcher to move recursively between the empirical data and theoretical concepts. This ensured a critical interpretation of students' understanding of speculative AI tools while navigating existing linguistic or other barriers. The process was divided into three distinct phases:

Phase 1: Familiarization, Manual Coding, and Analytical Memoing

The first phase involved initiation, manual writing, and analytical memorization. It focused on deep immersion in raw transcript files to ensure intimate familiarity with the nuances of each interview. This process began by reading the transcripts multiple times while listening to the raw audio to capture the emotional tone, misinterpretations, and linguistic distortions of non-native speakers. A manual open-coding process [18] was then conducted using printed copies of the transcripts. By physically highlighting "meaning units" on paper, an almost tactile connection to the raw data was preserved. To support this immersion, analytical memos were written for nearly every participant to document 'first impression' similarities or differences noted between participants, emergent patterns and reflexive thoughts. This step ensured that the social context of each interview was preserved before the data fragmented into smaller codes.

Phase 2: Spatial Synthesis, Participant-Anchored Mapping, and Iterative Refinement

The analysis transitioned to a digital workspace in FigJam for spatial mapping. Participant-Anchored Affinity Mapping was implemented, where each participant (P1-P14) was assigned a color-coded section. For each participant, the researcher simultaneously reviewed the transcript and re-listened to the raw audio recording, pausing at key moments to capture "meaning units" and significant reflections directly onto digital post-it notes. This real-time distillation ensured that the data represented a refined account of the participant's intent, capturing nuances in tone and emphasis that a text-only review might overlook. This process yielded a series of emergent codes that served as primary labels for describing specific student sentiments, such as *'Fear of Overreliance', 'AI as backup', 'Hesitant to ask Permission'*, and *'Bad at Multitasking'*. These codes acted as the granular data points that were later clustered into higher-level themes. Theoretical saturation was identified around the twelfth participant, as recurring patterns regarding linguistic load and social anxiety stabilized. Digital notes were reorganized from individual blocks into thematic "islands" based on conceptual similarity and iterative clustering. For instance, comparing the *"Conditions of Trust"* cluster with the *"Human Agency"* revealed their conceptual interdependence. Consequently, these were merged into a single, more robust theme that captured the negotiated relationship between the participant and the AI Notetaker Tool.

Phase 3: Thematic Categorization and Final Verification

Finally, these clusters were merged into a structured thematic framework. To maintain transparency, a *Thematic Comparison Table* was developed to outline core concepts, inclusion/exclusion criteria, and supporting participant patterns for each of the four final themes. (Appendix Figure 1)

Credibility was maintained through "re-hearing" cycles, verifying that the themes remained faithful to the participants' actual words. This systematic process provided a clear audit trail, moving the analysis from surface-level utility to an exploration of shifting classroom norms.

3.6 Ethical Considerations

The study adhered to the Swedish Research Council's ethical guidelines [12]. Informed consent was obtained digitally from all participants, who received detailed information and signed a digital consent form before participating. They were informed that participation was voluntary, that they could withdraw or skip any question, and that they could request to speak off the record without penalty.

To protect anonymity and confidentiality, all data were pseudonymized using alphanumeric codes (P1-P14), and any identifying details (e.g., names of specific courses or professors) were redacted during transcription. In compliance with GDPR, raw audio files were stored on a password-protected laptop accessible only to the researcher, and will be permanently deleted upon the completion of the thesis examination.

4 RESULT AND ANALYSIS

Students' engagement with lecture content revealed that note-taking is not a universal habit, but rather a highly individualized sensemaking process uniquely shaped by their personal habits, experiences, and academic needs. The majority of participants agreed that note-taking played an important role in their learning, but its functional value varied across the cohort, which manifested in 2 categories.

- Intensive Notetakers: Students who actively wrote during lectures, used notes as cognitive anchors, and relied on them heavily.

- Minimalist Notetakers: Students who prioritized active listening, took sparse or no notes, and relied primarily on post-lecture self-study.

The diverse engagement patterns described by students formed the contextual foundations for how they envisioned the future implementation of AI notetakers during lectures.

To navigate these varied perspectives, an analysis across 14 participants generated four overarching themes concerning students' perceptions of AI Notetakers in physical classroom settings:

1. *AI as Support, Not Replacement,*
2. *AI Notetakers as scaffolds for lecture comprehension, linguistic accessibility, and post-lecture Sensemaking,*
3. *Exploring responsibility, Classroom Dynamics, and the negotiated acceptability of AI Notetakers, and*
4. *AI as a Collaborator: Human Agency, Trust Calibration, and Acceptable Delegation in AI-supported learning*.

Across these themes, AI Notetakers were envisioned as collaborative, assistive, and bounded tools whose usefulness depends on their ability to support comprehension while preserving student agency.

4.1 Theme 1: AI as Support, Not Replacement

Students conceptualized AI Notetakers as a form of support to negotiate classroom tensions through cognitive offloading**,** specifically addressing the "split-attention" effect [45]. Envisioned through relational metaphors like *"AI Assistant"* or *"AI Friend,"* participants reflected a firm stance on the importance of not being replaced by technology, revealing a desire for a non-threatening, subordinate partnership tailored to students' convenience. P1 established a healthy boundary of power between human-AI interaction: *"I would love to use AI* Notetaker *as an assistant or AI Friend and not expect it becoming the student or replacing me."* This metaphor humanizes the technology while ensuring it remains peer-level support rather than an authoritative source of knowledge, which could erode the student's academic identity.

Similarly, P7 insisted on using AI as an assistive tool rather than a substitute for their note-taking, emphasizing they would use it only *"along with my writing"* to preserve handwriting as *"a feature that I do not want to lose."* This suggests that the acceptability of AI Notetakers relies on the tool's ability to remain an assistant, preserving the student's role as the primary meaning-maker. For these participants, note-taking is an *"embodied learning process"* where the physical act of writing facilitates concept-focused sensemaking through attentive listening, implying that manual recording remains essential to preserve intellectual agency [36].

4.1.1 The Tension of the "Backup" Mentality

Intensive notetakers valued their habit of taking notes; they identified struggles with the lecturer's pace or accent. They envisioned AI as *"absolutely useful"* for filling gaps and acting as a *"backup"***,** a term that implies a safety net rather than a primary driver. P5 pragmatically stated that the tool *"can definitely act as a backup source,"* rather than a replacement.

However, an abductive tension emerged, as some participants viewed this safety net as a potential disadvantage. P2 anticipated reduced classroom focus due to this *"backup mentality,"* noting: *"I may get distracted wisely thinking, oh, I have this backup".* This suggests moral licensing of distraction, as the existence of a backup makes the student feel they can let their mind wander. This sentiment, echoed by P10, P11, and P12, highlights that the presence of AI does not simply support students; it fundamentally reshapes their active cognitive engagement; transferring responsibility to a record-keeper can lead to shallow processing if the student believes their thinking is substituted by the tool [24, 25].

4.1.2 Autonomy and Fear of Laziness

The desire for structural support is countered by a significant concern regarding the erosion of cognitive identity and the fear of becoming passive observers. While students anticipated that AI could gather and structure notes to make them *"more digestible,"* particularly in dense and complex theoretical subjects, they expressed a prevalent fear that delegating the entire note-taking process might diminish their active presence. This tension manifests not just as a loss of notes but as a potential undermining of classroom engagement, leading to a sense of *"I don't have to be there"*. Unpacking this sentiment reveals that "*being there"* is viewed as an active intellectual contract that automation threatens to break.

This anxiety is articulated through a recurring fear of laziness and overreliance. For instance, P8 emphasized a desire to maintain student autonomy, stating: *"If I have this kind of AI tool, I will become lazy."* This was echoed by P11, who cautioned that AI Notetakers might induce a feeling of *"lethargy"* if not used mindfully: *"Maybe it can make you a bit lazy...if you are not using it responsibly."* In an academic context, this laziness is not merely about physical effort; it is a fear of cognitive offloading becoming so extreme that the student is no longer the primary thinker in the room. This tension was particularly pronounced among intensive notetakers, who feared that AI support would erode the "desirable difficulties" of learning [42]**.** Consequently, if the boundary between a student's intellectual contribution and an AI's automated function becomes blurred, these tools risk hampering critical thinking and fostering cognitive lethargy unless they are explicitly designed and taught for responsible use.

4.2 Theme 2: AI Notetakers as Scaffolds for Lecture Comprehension, Linguistic Accessibility, and Post-Lecture Sensemaking

AI Notetakers are valued as tools for helping students navigate the complex temporal flow of a lecture. For these students, perceptions of AI were shaped by their specific linguistic positioning and prior educational exposure. While they wanted the tool to be non-intrusive, they primarily viewed it as a scaffold: a necessary support to reach a level of understanding that English-medium instruction often makes difficult to achieve alone.

4.2.1 Bridge for Language Barriers: From Exhaustion to Confidence

Language emerged as an important but uneven theme. Several participants with English as a secondary MoI described lectures in English as a significant struggle, where a professor's pace or accent could disrupt real-time comprehension. This reality directly reflects the hidden linguistic burden of English-Medium Instruction (EMI) [30] rooted in Cognitive Load Theory [38]. For these individuals, note-taking involves an intense "translation layer" spanning translating, comprehending, and synthesizing [36]. This compounded effort leads to the exhaustion voiced by P6: *"I had to first think in native and then English and then write. It was very tiring…I felt that the two hours were four hours."*

To mitigate this exhaustion, the AI Notetaker was envisioned as a compensatory support. P1 also highlighted a lack of confidence in her accuracy: *"English is not my mother language, so most of the times, I am not sure if I take a note correctly or not, and therefore I need to check it with AI."* This reliance on AI as a validation partner reflects the need for linguistic scaffolding and psychological reassurance [48].

Conversely, participants with substantial prior experience in EMI environments did not describe language as a major barrier —P10, P11, P12, and P13 felt more comfortable with academic English, framing the AI Notetaker as a convenience tool to optimize studying rather than an absolute necessity. Interestingly, 2 minimalist notetakers, despite having English MoI, expressed a desire to delegate transcription entirely, viewing the tool as a convenience. However, an abductive tension emerged between P1 and P8; despite facing similar linguistic barriers, P8 responded more cautiously, fearing that the tool might shift participation away from the student and toward the device itself**.** This highlights a pedagogical limit of AI [42], where the benefit of support must be weighed against the risk of losing student autonomy.

Collectively, these findings suggest that while language barriers increased the perceived relevance of AI Notetakers for some, they did not automatically lead to positive acceptance, demonstrating that perceptions varied significantly even among students navigating with similar linguistic difficulties.

4.2.2 Post-Lecture Sensemaking: Transforming Data into Knowledge

Beyond real-time assistance, six participants envisioned AI Notetakers as retrospective post-lecture sensemaking tools [49], where meaning is constructed after the lecture[18, 30]. For several participants, this meant transforming verbatim text into visually and relationally meaningful forms. Participants demonstrated a pronounced desire for structural affordances that extend beyond linear text, conceptualizing the AI tool not merely as a transcriber but as a generator of relational mind maps.

Rather than requesting a specific feature for its own sake, participants expressed a cognitive need for visual synthesis to help them manage the *"overwhelming"* nature of full-text records. This demand aligns with frameworks demonstrating that visual diagrams reduce cognitive load by externalizing thoughts into spatial relationships compatible with human mental models [46].

This orientation highlights a shift toward AI-mediated sensemaking [44], where the tool serves as a bridge between fragmented lecture data and a cohesive mental model. This is exemplified by P9's vision of a *"Dashboard View"* that represents a longitudinal sensemaking mechanism, implying not just an interface preference, but a way of connecting previous lectures to encourage retrospective understanding. By providing a *"bird's-eye-view,"* the AI is envisioned as a structural guide that assists in navigating dense content and identifying conceptual threads over time, as suggested by Bui et al. [6].

Finally, the desire for a minimalist and non-intrusive interface across the cohort suggests that while students want powerful visual tools, they want them "*backgrounded*" during the lecture itself. This aligns with Norman's principles of unobtrusive design, where a tool remains discoverable but does not compete for the user's limited attention during the primary activity of listening [31].

4.3 Theme 3: Exploring Responsibility, Classroom Dynamics, and the Negotiated Acceptability of AI Notetakers

The social legitimacy of AI Notetakers is fundamentally defined by established classroom norms, collective social expectations, and evolving frameworks for consent regarding permissible recording. Throughout the study, participants identified a shared landscape of responsibility: while students recognized their own role in the ethical use of the tool, they also maintained clear expectations for institutional oversight and teacher-led boundary-setting. Three subthemes explore the hierarchy of permission, the panoptic effect on classroom behavior, and concerns around classroom engagement.

4.3.1 The Hierarchy of Permission and the Responsibility Gap

Seeking permission emerged as a primary ethical prerequisite before introducing an AI device in the classroom. P3 expressed immediate concern regarding the ethical recording of teachers and peers without explicit consent, fearing social stigmatization as a *"defaulter"* who breaks classroom norms. P6 echoed this sentiment, framing consent as a matter of respecting educators' efforts to prepare their materials; *"They have the right to be asked if they want to be recorded or not...they have put a lot of effort to create their notes and their slides and everything."* Bypassing this preparation was described as *"unfair"*, indicating that for many, the recording tension leads directly to the requirement of formal consent.

However, a contradiction emerged with P8, who stated they *"did not feel that permission was as important",* viewing professors more as public figures, and assuming that recording them was less problematic. Despite this, a significant "Responsibility Gap" was observed: almost all participants were hesitant to ask for permission themselves. This hesitancy was exemplified by P1, who expected professors to explicitly state restrictions if they had objections. This mindset suggests an "opt-out" mental model, where students assume the absence of restriction as implicit permission— a behavior that aligns with privacy defaults research, where missing restrictions are misread as implicit permission and a signal of the established social norm [1]. To bridge this gap, P4, P5, and P12 advocated for the requirement of institutional policies and educating on ethical use, aligning with frameworks that suggest controllable systems succeed when the user's active role is strengthened [28].

4.3.2 Classroom Ecology: Negotiating Privacy, Surveillance, and Engagement

AI Notetakers are not envisioned as "invisible" tools, but as active agents that reshape the classroom ecology. P7's account positioned consent as central here, emphasizing the necessity of *"respecting collective privacy"* in shared educational spaces. P7 anticipated that if classmates recorded without consent, it would impact behavior, *"I will not ask questions,"* demonstrating that acceptability is about shared agreement, transparency, and mutual respect for consent.

This anticipated behavioral shift closely mirrors Foucault's Panopticon—a surveillance model where the awareness of being recorded induces self-regulating behavior [13, 17]. P1, P5, and P6 noted that the presence of an active recording tool, could make students self-conscious and anxious. For a few, the '*fear of being permanently captured*' can create a distinct "Chilling Effect" [17], potentially prompting them to stay silent rather than engage. While other students reported they would remain comfortable or neutral being recorded, the prevailing trend indicates that the classroom environment is highly sensitive to permanent capture, risking spontaneous engagement. To mitigate these speculative surveillance risks, participants expected the tool to function as an *"Intelligent Notetaker"* that filters out *"off-topic student chatter,"* ensuring the final output is a clean, ethically safe record. This expectation suggests that a technical safeguard could protect collective privacy and restore a sense of conversational safety in future classrooms.

Beyond these privacy concerns, this altered ecology introduces a parallel tension regarding lecture attendance and active attentional effort, as articulated by P5, P6, P9, and P14. A prevalent fear exists that if AI-generated notes are seen as "sufficient", it might encourage *"lecture skipping"* and lower the motivation to physically attend lectures. To counteract the fears of passive reliance and student disengagement, P11 suggested an interactive "*quiz feature"* where students must correctly answer lecture-related questions to unlock access to the full AI-generated notes.

Ultimately, these design expectations demonstrate that the legitimacy of AI Notetakers depends on their ability to preserve collective privacy while enforcing active student engagement.

4.4 Theme 4 AI as a Collaborator —Human Agency, Trust Calibration, and Acceptable Delegation in AI-Supported Learning

In this theme, the AI Notetaker is framed as a collaborator that allows students to actively intervene and retain control. This framing emerges directly from participants' own words, relational metaphors such as '*assistant*,' '*friend*,' and '*collaborator'* and their repeated insistence on remaining the primary meaning-maker in the learning process. Participants viewed preserving agency over their notes as an immensely important aspect. Notably, this desire for customization was present across both intensive and minimalist notetakers, suggesting that agency over output format is a universal student concern regardless of existing note-taking habits. Four subthemes address customization, the risk of dependency, conditional trust, and epistemic boundaries of acceptable delegation.

4.4.1 Customization as a Condition for Usefulness

When asked about their structural preferences for an AI Notetaker, the responses were quite interesting and recurring. No student wished for AI notes to be fully autonomous or locked. As P2 states: *"I think it's always nice to have the opportunity to add your own thing somewhere"* who views the ideal AI Notetaker tool as a collaborator and expresses*: "it's important that you are able to have agency, like you have the possibility to change things or to structure it the way you want. You are part of it. It's not just a tool. It's something that collaborates with you. And that makes a difference."* P3 similarly favored personalized outputs, noting: *"I wouldn't be that happy about fixed notes because sometimes I like to write down something like my own, which makes notes mine like funny comments."*

These preferences indicate that students valued control, customization, and optionality. For instance, P1, P4, and P6 expected functionalities that permit *slide screenshot uploads*, as combining visual components with text directly assists their learning style and strategies [11]. Structurally, the cohort preferred bullet points and concise texts, while discouraging text-heavy notes. Furthermore, highlighting key concepts was a near-universal expectation reflecting students' strong desire to preserve their existing note-taking habits. Additionally, three participants specifically wished for references to the taught content, reflecting a desire for validated sources over random information. This indicates that students require features that allow them to proofread and cross-check the AI's output to have confidence and build trust.

The overwhelming desire for customization and control reflects a fundamental student need to preserve their "first-person sense of agency" (SoAFP) and autonomy in their learning process**.** The cohort's insistence on editing and restructuring information (*mind maps*) aligns with Self-Determination Theory, which highlights autonomy as a core psychological need for intrinsic motivation. In HCI, this desire for diverse ways to organize information connects to the principle of "equifinality"—the concept that users feel significantly more empowered when they have multiple pathways to achieve their desired outcome.

Systems that allow users to act independently and exert fine-grained control enhance their sense of agency, preventing them from feeling subordinate to the AI [28].
Using sensemaking theory, this suggests that students view AI Notetakers not as authoritative tools that dictate their learning and actions but as mere generators of accurate raw materials that must be actively molded to fit their personal cognitive schemas.

4.4.2 Preserving Agency Against Dependency

This subtheme captures participants' concern that while AI Notetakers may reduce cognitive burden, they may also encourage overreliance, reduced attentional effort, and diminished classroom engagement. P2 emphasized that the AI tool should position itself as a collaborative partner :*"Let's do this together, collaborate",* as this will allow students to add their own knowledge and expertise into their notes, suggesting a framework for *"Human-AI co-note-taking,"* where both agents engage in collaborative sensemaking.

Participants expressed substantial fear of becoming dependent on AI tools, viewing overreliance as a risk to their learning autonomy. So, preserving their own identity and deciding which tasks to delegate to AI reflects an active exercise in critical thinking and awareness of not becoming overly dependent on these tools. Prior studies have highlighted a significant paradox regarding educational technology: tools designed to enhance learning may, if over-utilized, simultaneously undermine the very skills they are intended to support [40, 42].This tension is captured in P10's reflection: *"I hope that people don't become too dependent on it, but I would be a bit more sure that even if I don't pay 100% attention during the lecture... I'll still get all the information"* which, shows the beginning of dependency logic where cognitive reassurance can be helpful, but can unintentionally reduce attentional effort. These accounts suggest that participants evaluated AI Notetakers not only in terms of convenience or efficiency but also in terms of their potential effects on students' own engagement and responsibility. AI support was seen as valuable, but only if it did not erode active participation, attention, or the learner's role in constructing understanding.
To preserve agency, all participants expected this human-in-the-loop to be a core feature. Integrating AI Notetakers requires establishing a productive "epistemic relationship" between the student and the technology [51]. Ideally, students must engage in *Co-agency Collaboration,* treating the AI as a thinking partner while remaining the primary builder of knowledge. Without this critical balance, excessive reliance poses a significant threat to student agency as overreliance risks inducing "solution paralysis"—an inability to initiate problem-solving without technological assistance [40].

4.4.3 Trust as Conditional and Continuously Calibrated

Participants did not describe trust as a static attribute of the technology, but rather as a dynamic, continuous calibration process re-evaluated based on the AI's ongoing accuracy. Trust is not calibrated automatically; instead, it is conditionally calibrated based on the system's accuracy and its ability to allow students to have agency. For several participants, trust was not automatic; instead, it was anticipated to develop based on its accuracy. When asked if they would trust tool without review, P6: *"No… I'm building LLMs so I know, I mean, I see that they are hallucinating a lot, they would never say they don't know a specific thing."* P6 described their current stance as being *"in between trusting completely and trusting not at all."* Similarly, P4 noted *"I don't completely trust it to get everything right,"* while P5, drawing on prior experiences with incorrect AI responses, remained skeptical, and trusted their own notes.

These accounts demonstrate that, because AI models are prone to hallucination, trust development is hampered. Beyond accuracy, participants like P11 expressed concern regarding data privacy and *"what happens to the data that it records,"* highlighting that even functional trust is often tempered by curiosity or caution regarding data usage. For many participants, this conditional trust was rooted in prior interactions with AI tools that produced inaccurate, misleading, or overly confident outputs. Consequently, participants adopted a stance of cautious use: leveraging AI output for reminders or missed content, while insisting on verification against what was actually taught. Trust is therefore granted only provisionally and must be supported through review, verification, and cross-checking.

This orientation is clearly reflected in P10's account: *"I don't think I would trust it completely ever,"* indicating that trust in the AI notetaker would remain inherently partial; even when the tool might successfully capture missed content, participants emphasized the need to *"cross-reference and make sure everything is correct."* Collectively, these perspectives suggest that

participants positioned themselves, rather than the AI, as the final arbiters of accuracy and relevance, imagining AI Notetakers less as trustworthy authorities and more as provisional collaborators whose outputs remain subject to human validation.

4.4.4 Acceptable delegation and epistemic boundaries

A core challenge in AI-supported learning is defining the boundaries of which tasks can be appropriately delegated to AI. Participants drew clear lines between clerical assistance and interpretative work, distinguishing between forms of support that facilitate learning and automated interventions that risk overstepping into misinterpretation or distortion. Participants expressed more comfort delegating transcription, information retrieval, and layout structuring than deeper interpretive tasks like filling in missing information. Students explicitly expect the tool to remain tethered to what was actually said, rather than attempting to repair uncertainty through algorithmic guesswork.

P10's account illustrates this epistemic boundary clearly: *"I would want something that's quite one-to-one… I'd much prefer if it just leaves it blank,"* rather than filling in missed content. Similarly, P11 noted that if the tool misses information, it should **"***not fill in with misinterpretation, instead keep a gap".* These insights suggest that students prioritize transparency over fabricated coherence (i.e., hallucination), especially when preparing for high-stakes evaluations. This reveals a strict epistemic boundary: to a student, navigating a *"blank space"* is far more acceptable than relying on invented text implying uncertainty is more acceptable than hallucination. In this sense, participants' openness to AI Notetakers is heavily bounded by epistemic expectations regarding data integrity.

Furthermore, acceptable delegation was shaped by operational conditions; five participants linked the tool's academic legitimacy to privacy protection, localized storage, a neutral tone, and automated filtering of extraneous student conversations to ensure the output remains clean and ethical. Crucially, the absence of emojis was framed as an essential factor for academic acceptability, reflecting a clear expectation that AI-generated notes must serve as a disciplined, scholarly record of information to be trusted. Ultimately, participants endorsed a framework where AI restricts its role to structural capture and organization, leaving the interpretation of meaning safely within human boundaries to respect and support active student engagement[10].

5 DISCUSSION

5.1 Perception and Anticipated Roles of AI Notetakers

This study investigated how university students perceive and anticipate the role of AI notetakers in physical classroom environments. The findings indicate a nuanced ambivalence, rather than straightforward acceptance or rejection. This thesis argues that students anticipate accepting AI notetakers not as substitutes for their engagement but as identity collaborators and learning scaffolds, whose acceptability depends on the extent to which they preserve student agency, support active learning, and respect the social norms of shared classroom spaces. More broadly, this study questions about the implications that can be brought by the introduction of AI notetakers in classrooms, who has the agency, and how the presence of such a tool might reshape the classroom dynamics.

Interpreted through Weick's sensemaking[49] lens, the relational metaphors that students assigned to AI notetakers like assistant, friend, collaborator – are not arbitrary choices; rather, these reflect a deeper identity-construction process through which students established a plausible and psychologically safe relationship with an unfamiliar technology.

The analysis revealed that linguistic background played a significant role in shaping how students constructed the tool's potential value — a dimension that the retrospective property of sensemaking helps explain.

5.2 Expectations and Benefits

Students anticipated that the presence of AI Notetakers in the classroom could redefine active learning and impact engagement, suggesting these tools are envisioned not as mere invisible tools but rather as visible collaborators. This shift is deeply tied to identity construction [49], as participants perceive the classroom as the primary stage where their identity as "students" is performed and maintained. Relational metaphors like *assistant, friend,* or *collaborator* reveal that students humanize technology to negotiate a hierarchy that preserves their identity as the primary meaning-makers. Because this identity is inextricably linked to attending lectures, the anticipated introduction of AI was envisioned as a potential reconfiguration of

their role. Students strive to preserve active engagement and agency through human-in-the-loop systems while navigating the split-attention effect [45]. This suggests that for AI Notetakers to be perceived by students as socially and pedagogically acceptable, they would need to value and respect students' intellectual agency.

The distinction based on MoI shifts the perceived utility of AI Notetaker from to a necessity to a convenience. Students with secondary EMI backgrounds anticipated positive behavioral changes, such as increased confidence in their understanding of the content taught in class, highlighting a critical shift in the tool's purpose; students envisioned it evolving from a record-keeper to an affective scaffold. They envisioned it would help reduce the cognitive load of linguistic barriers, thereby allowing them to focus on active listening. This moves the tool beyond a convenience to become a comprehension support. However, the backup mentality identified in the findings, when interpreted through the lens of cognitive offloading [38], introduces a paradox of passive activity. In the broader sense, students' accounts suggest that their presence in class would no longer ensure active learning, as they may prioritize supervising the performance of AI Notetaker over the effortful deep processing required for sensemaking.

This finding is rooted in the retrospective and extracted cues aspects of sensemaking [44, 49]. Students draw on past experiences of "*zoning out"* to frame AI as a bridge for moments of inattention. Interestingly, the widespread desire for mind maps and connections validates the retrospective nature of sensemaking, suggesting students require spatial organization of lecture content to manage complexity. By transforming linear text into visual connections, these tools align with the concept of equifinality [28], providing multiple pathways to reach coherent knowledge [46].

Finally, a clear expectation from the tool across the cohort was minimalism and non-intrusiveness, such as non-competing icon reveals students' acute awareness of classroom norms and the intrinsic value of being present. The preference for an *"intelligent notetaker"* that filters student conversations and respects social boundaries suggests that students are not looking for a tool that replaces the lecture experience instead one that respects the traditional boundaries of teacher-student interaction.

The widespread preference for minimalism suggests that student expectations are increasingly shaped by broader trends in Human-Computer Interaction (HCI), where Calm Technology [50] with unobtrusive design has become a standard. As users become habitualized to minimalist AI interfaces in their digital lives, they carry these expectations into educational domains [31]. Consequently, the demand for a non-intrusive tool reflects the desire for technology that encalms the user and assists the learning process without disrupting the classroom ecology [29]. In an era of notification fatigue, participants expressed that introducing tools that compete for the same cognitive resources required for active listening and engagement is nonsensical.

5.3 Concerns, Challenges, and Acceptability of AI Notetakers

Concerns regarding consent and classroom norms disruptions reflect what sensemaking theory describes as a *collective meaning-making* process [49]. Rather than making individual judgments, students envisioned a collective negotiation and the social legitimacy of recording devices within an established classroom. This indicates that sensemaking for AI Notetakers is an inherently social process in which students interpret their surroundings amid ongoing change. Students anticipated that the introduction of a recording device like an AI Notetaker into shared classroom spaces could disrupt established norms, emphasizing the need for consent and collective agreement between educators and students.

The anticipated presence of a recording device elicits a surveillance tension among students, where the fear of being captured leads to a self-regulatory hesitation to engage openly in class discussions. Interpreted through Foucault's theory of surveillance[13] the tool could risk transforming the classroom into a space of judgment, where students might self-censor to avoid being captured permanently. The acceptance of this tool was not uniform among students, as it was shaped by prominent concerns regarding trust, and epistemic safety, reflecting sensemaking's ongoing nature, as acceptance was never fixed but perpetually conditional.

This conditional acceptance is further complicated by existing power dynamics, as students expressed a clear expectation for explicit permission from teacher. Students' hesitation to ask for permission themselves suggests that they view the teacher as the primary authority maintaining an ethical atmosphere in the classroom. This desire for institutional scaffolding [44, 49], where policy and consent are handled at a higher level, reveals that students do not want to bear the social burden of implementing the tool on their own. Additionally, the fear of overreliance on the tool signifies a potential threat to student autonomy, a core psychological need under Self-Determination Theory [28]. Students expressed concern that dependency on AI notetakers could normalize patterns of passive learning, potentially eroding their identity as independent learners.

A further concern among students centered on AI hallucinations; participants expressed a preference for "blank gaps" over "fake information", emphasizing the need for agency and accuracy. Taken together, these concerns suggest that the challenges of implementing AI Notetakers in classrooms are not purely shaped by the technicalities of the tool; instead, they are entangled with deeper concerns about identity, autonomy, and the long-term preservation of students' roles as active learners. Collectively, the concerns highlighted by students indicate that acceptance of such novel technologies in shared spaces requires attention to the social, ethical, and motivational aspects, thereby providing a safe environment for learning and engagement without hesitation.

Significantly, students do not reject the idea of implementing AI Notetakers provided it stays within ethical boundaries. By viewing teachers and institutions as figures of higher-order authority, students acknowledge the social legitimacy of AI in shared spaces requires a formal set of rules and regulations. This reflects that understanding of responsible AI use in shared spaces requires institutional scaffolding, not just individual goodwill. This positions students not as passive recipients of technology but active negotiators in deciding the terms under which such tools become ethically beneficial and acceptable.

Design Implications

This section outlines future design implications for designers, HCI researchers, and other scholars. The findings highlight that students expected AI notetakers to move beyond basic information-capture devices to support a diverse spectrum of student engagement. To serve both intensive and minimalist note-taking preferences, design interventions should emphasize customization, user agency, transparency, and cognitive scaffolding.

1. **Support dynamic agency**: The design of the tool should incorporate students' preferences, allowing customization for the structure, information, and also support dynamic agency by enabling a selective capture of important details from a lecture by providing them with real-time highlighting.
2. **Scaffolding for Linguistic Load**: To support NNE students, features such as real-time definitions of complex terminology can reduce cognitive overload without disrupting their engagement patterns.
3. **Epistemic transparency**: Design should prioritize accurate and verifiable capture and leave blank gaps over hallucinated content or mark the content as tentative so that the student knows where further verification is needed. Reliability indicators and source references should be embedded to indicate the source of the information and verify its authenticity, enabling students to independently verify the accuracy of captured content.
4. **Visual aid for post-sensemaking**: Crucially, the tool should be designed not just for real-time lecture capture but as a post-lecture sensemaking scaffold that offers structured summaries, dashboard views connecting across lectures, and visual features such as mind maps that support retrospective understanding.
5. **Gamified access**: Perhaps the most original design insight emerged from one participant, who proposed an AI notes-unlock mechanism in which AI-generated notes would only become accessible once the student submits a summary or a quiz of what they had learned, which the system would compare against captured content and grant access accordingly. This reflects the enactment principle of sensemaking, where active participation in the process sustains the student's role as the primary meaning-maker.

By shifting from *'automation' to 'augmentation,'* AI notetakers can potentially serve as collaborative partners that respect individual learning habits while minimizing the cognitive and social challenges often found in lecture environments.

5.4 Limitations & Future Directions

This study presents several limitations and strengths that merit consideration. The qualitative sample of 14 participants, though sufficient for exploratory studies and appropriate for achieving theoretical depth in the subject, is not intended to generalize beyond the study's specific context. Findings were theoretically generative but not statistically representative. Additionally, this study was conducted before the actual implementation of AI Notetaker in classrooms which means, it captures students' imagined use of speculative technology, rather than interaction with live prototype. While acknowledged as a limitation, this approach is a methodological strength, as user studies in HCI frequently precede the actual design implementation to understand user needs, pain points, and expectations, thereby informing prototype development.

As a sole researcher, I acknowledge the risk of interpretive bias during thematic analysis. To mitigate this, I employed an abductive analysis, maintaining a reflexive engagement with the data throughout the research process. Additionally, because

all 14 participants were NNEs, these findings reflect experiences of international student population rather than native speakers, suggesting a valuable direction for future research.

Finally, the classroom comprises both students and teachers; however, this study focused exclusively on student perceptions. Although teachers are an integral part of the classroom ecology, their perspectives remained peripheral due to scope constraints. It was a deliberate choice to focus on students due to scope constraints, but this opens the door for conducting future research with teachers on this subject, as a dedicated study foregrounding teacher perspectives would offer a more complete picture of classroom-level acceptability and institutional readiness. The design implications generated by this study should therefore be understood as informed directions, grounded in student voices rather than validated prescriptions, and require iterative prototyping and testing with broader populations before implementation.

6 CONCLUSION

This study explored how university students perceive and anticipate the role of AI Notetakers in physical classroom settings. The findings reveal that the roles of these utilities are carefully negotiated and conditionally shaped by concerns about identity, agency, and social responsibility. Across all themes, students evaluated AI Notetakers not as mere technical tools but as active social elements envisioned either as supportive learning scaffolds or collaborators within the classroom ecology. By applying Weick's Sensemaking Theory, it became evident that students envisioned evaluating the presence of these devices based on how well the technology integrates their expectations and concerns. Importantly, students demonstrated a critically aware mindset—recognizing the importance of institutional scaffolding, collective consent, and responsible usage in determining ethical integration. They positioned educators and institutional bodies as pivotal in deciding the rules and regulations of its implementation. In the emerging era of AI in the classroom, it is crucial to adhere to students' expectations and classroom norms to ensure a safe environment for knowledge transfer.

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AI Declaration: I declare that Gemini, Claude, and Grammarly were utilized strictly as writing aids to help with language editing, proofreading, grammar corrections, and improving the overall flow of text. These tools were used exclusively to refine the academic tone and polish clarity of drafts that I wrote by myself. No data, qualitative analysis, or core arguments were generated by AI. This thesis remains completely my own work, and I take full responsibility for its final integrity.

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9 APPENDICES

INFORMATION SHEET

My name is Priyashi Dogra, a current 2nd year Masters Student in Human-Computer Interaction at Uppsala University. I would like to ask you, if you will like to participate in this study which is conducted as a part of my thesis: *Exploring the role of AI Notetakers in Physical Classrooms Settings: An HCI Study*. In this document, you will receive information about the project and what it means to participate in this study.

What does the Interview involve?

The interview will be more of a guided conversation which has no right or wrong responses. The aim of this interaction will be to know your perceptions about AI tools usage in classroom specifically a notetaking tool. The only prerequisite that is required is that participant must have AI tools exposure and should have experience of attending physical lectures in English.

What happens to your data?

The collected data is stored in a password protected secure device which can be accessed only by the researcher. The interview will be text transcribed using a GDPR compliant tool for analysis purpose and will be deleted immediately after converting to text transcription. Once the thesis is defended all the data will be deleted thereafter.

Purpose

The purpose of collecting and processing of personal data is to conduct academic research for my Master Thesis. The personal data that will be collected from you will include basic demographics like name, program of study, age. All the data will be anonymized during transcription, no direct identifying information will be included in the final thesis report. Your data will be processed for duration of the study only after which it will be deleted permanently. All personal data will be processed in accordance with the requirements of GDPR.

The interview will take around 30 to 45 minutes and will involve general questions around your class engagement patterns, AI tools usage, notetaking preferences, struggles and imaginary based questions for a future based AI Notetaker tool.

Participation Information

- The participation in this study is completely voluntary which implies you can withdraw anytime from the studies.
- You can also request to delete your data before the thesis completion if you wish to do so.
- The interview will be recorded and the data will be dealt as per GDPR compliance.

CONSENT FORM

By signing this consent form you agree to process your personal data for the purpose of this thesis study. Your consent constitutes the legal basis for the processing of your personal data. Your participation is completely voluntary, and you can withdraw your consent at any time and request a copy of your personal data. If you wish not to reveal your identity, you can fill in 'Anonymous.'

1. Name of Participant:

---

2. Pls choose your Age Group:

- 20-25
- 25-30
- 30 and Above

3. Degree Program Enrolment:

- Bachelor's Degree
- Master's Degree

4. Program Name:

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5. Consent for Recording and Data Use :

- ☐ Audio Recording: I consent to this interview being audio recorded for the purpose of accurate transcription.
- ☐ Anonymity Preference: I wish to remain anonymous in the final report (my name will be replaced with a pseudonym).
- ☐ General Consent: I confirm that I have read the information above and voluntarily agree to participate in this study.

6. Participant's Sign:

Researcher: Priyashi Dogra

Contact Detail: priyashi.dogra.3166@student.uu.se



INTERVIEW GUIDE

This study is conducted for a master's thesis where I am exploring students' current note-taking practices and their perspectives on how they would perceive AI note-takers or AI note-taking tools in physical classroom settings. There are no right or wrong answers. Whatever your experience, perceptions, and viewpoints are, I'm interested in knowing them. And if you feel you don't have an answer, you can just skip the question. You're totally free to do so. The data I am collecting will be used for the thesis and then deleted.

Section 1: Background & Study Context, AI Familiarity (12 mins)

Warm-up questions

1. Tell me about yourself.
2. Could you please describe your current study program and the types of lectures you attend?
3. How do you engage during lectures in the classroom?
    - Probe: Do u mostly listen, take notes, ask questions, discuss with peers?
4. How do you usually take notes during lectures?
5. Have you used any AI tools for studying or academic work?

→ Have you ever used an AI-based note-taking tool?

Section 2: Cognitive & Classroom Experience (15-20 mins)

6. What role does note-taking play in how you learn during lectures?
7. What do you typically do with your notes after class?
8. During lectures, do you ever struggle to keep up — whether because of speed, complexity, or language? / Can you describe situations in lectures where it becomes difficult to keep up with the content?

Follow-up:

- Does studying in English influence your experience in any way?

9. Can you recall a recent lecture where taking notes felt particularly difficult?
10. How do you decide what is worth writing down?
11. How do you usually handle unfamiliar terminology/words or fast explanations?
12. In what ways does language influence your concentration or comprehension in class?

Section 3: Introducing the AI Notetaker (Core Reflection) (15 mins)

Now I'd like you to

"Imagine an AI-based note-taking tool that runs during your lecture. It listens to the lecture (using speech recognition), automatically transcribes it, and then uses AI to generate structured summaries and key points. After class, you can ask it questions about what was discussed, and it responds based on the lecture content."

Then:

10. What is your immediate reaction to this idea? What role do you think it should play in your student life?
11. What are your expectations from this kind of AI tool?
12. How do you think this tool would change your behavior during lectures?
13. Would you still take your own notes? Why or why not?
14. In what situations could this tool genuinely support your learning?
15. In what situations might it reduce engagement or create dependency?

Section 4: Agency, Ownership & Learning (10–12 mins)

15. How would you prefer to interact with the AI-generated notes to make them most useful for your learning?? (fixed static summaries/actively edit or reorganize)

    Probe: editing, reorganizing and adding personal notes

16. What would make the notes feel like your own rather than something produced by the AI?
17. How would AI-generated notes fit into the way you currently review or personalize your notes?
18. Would you trust AI-generated notes without reviewing them? Why or why not?

Section 5: Course Suitability & Boundaries (3 mins)

19. In what types of courses do you think this tool would work particularly well?
20. Where might it not work?

Section 6: **Impact on Lecture Experience (12 min)**

21. If this tool were running during your lecture, how do you think it would change what you do in class?
22. You zone out for 10 minutes, daydreaming. The AI captures those 10 minutes perfectly. But you have no context for them. Are those notes useful or confusing?"
23. "If you asked a question that seemed 'stupid' in class, and you knew five classmates' AI tools captured it permanently, would you still ask it? Why or why not?"
24. How might this tool affect your attention or concentration during lectures?
25. How big of a problem will it be if this AI tool would miss some information or makes mistakes?

Section 7: Responsible & Acceptable Use (10–12 mins)

26. *How do you think the use of AI notetakers by some students might influence other students in the class?*

27. If AI tools capture everything said in class, do you think that could create new responsibilities or concerns for students?
28. How would you feel about an AI system recording lectures in a physical classroom?

Follow-Up Probes (Use Only If Needed)

If participants stop talking, I will gently probe:

- "Are there any concerns that come to mind?"
- "How might this affect students or teachers?"
- "Would permission from the teacher make any difference?"
- "What would make this feel acceptable in a classroom setting?"

Theme	Description/Concepts	Includes	Excludes	Supporting Codes	Participant Patterns
Theme 1: AI as support, not replacement	Students perceive AI note-takers as supplementary/assistive tools that support learning, rather than substitutes for attention, understanding, or note-taking responsibility.	AI as backup, passive support, safety net, "just in case" tool, support for remembering missed content, not relying on AI completely	Specific feature expectations, privacy concerns, classroom consent, customization/ control	• AI as a backup for missed meaning • Sensemaking is concept focused • AI as reassurance when attention is imperfect • AI as support, not replacement • overreliance; • Risk of laziness • student losing autonomy • Fear of overreliance • Skip lectures • Stop taking active notes • Note-taking as personal learning process • Embodied cognition / learning through writing • Resistance to outsourcing thinking • AI as assistant, not proxy/ substitute • Fear of losing a valued skill • Preservation of authorship • Personal ownership of notes	Participants described AI notetakers as helpful / useful tools for recovering missed content from lecture, for accuracy but not as something that they would not overrelai or depend on as that would make them lazy and passive learner, some participants recognized it could also harm their critical thinking
Theme 2: AI Notetakers as scaffolds for lecture comprehension, linguistic accessibility and post-lecture sensemaking	AI note-takers are valued for helping students follow lectures, overcome language-related barriers, revisit content, and organize understanding across time.	Pace/speed difficulties, accent and tone issues, English-medium lecture challenges, post-lecture review, revision for exams/assignments, visual features such as mind maps, dashboards, overviews, minimalist interface, support for dense/technical content	Privacy/storage issues, classroom legitimacy, manual control unless tied to usefulness, hallucination/inaccuracy as a risk	• keeping up with pace; • accent and tone barriers • post-lecture review • Hard to Comprehend in english • Thinks in their native language • Multitasking is difficult • Gets fixated on phrases • Attention span limitation • post-lecture support • visual sensemaking support • mind maps for conceptual linking • bird's-eye-view overview;	Participants with less prior English-medium study saw stronger value in AI note-takers for handling accent, pace, and comprehension; others saw them more as convenience tools.
Theme 3: Responsibility, classroom dynamics, and the negotiated acceptability of AI note-takers	The acceptability of AI note-takers is shaped by classroom norms, social expectations, and negotiated ideas of consent, permission, and appropriate recording.	Permission, explicit vs implicit consent, whether teachers should set rules, recording teacher voice, recording classmates' questions, off-topic chatter, classroom legitimacy of AI recording	Technical accuracy concerns, privacy concerns unrelated to classroom use, customization, general usefulness	• classroom dynamics • consent • responsibility to inform • recording without permission • off-topic chatter • self-conscious students • hesitant to speak • Attendance concerns • Responsible use	Some participants assumed permission unless prohibited, while others emphasized the need for clearer consent and boundaries around recording in classrooms.
Theme 4: AI as a collaborator: human agency, trust calibration Accepted Delegation in AI-supported learning	Students want AI note-takers to remain under human control, with trust granted conditionally through editing, customization, and verification and clear limits on what may be delegated to AI.	Customization, editing, restructuring, highlighting, cross-checking, reviewing outputs, dependency concerns, AI as collaborator rather than authority, preserving learner agency	General usefulness statements, privacy/ consent issues, language-related benefit, technical concerns unless tied to user control	• Trust through verification • Fear of AI hallucination • Conditional delegation • Moral limits of automation • Customization • Preserving student attentiveness • Accountability through forced participation • Conditional access to notes • Accuracy concerns • Context preservation • Student control • Student agency • Personalization needs • Human-AI co-note-taking	Across participants, customization was consistently valued, and AI was imagined as something to work with rather than surrender learning to.
	Students' willingness to rely on AI depends on accuracy, privacy, epistemic restraint, and limits on how far AI should interpret or generate content.	Hallucination, inaccuracy, misinterpretation, wrong definitions, exam risk, one-to-one transcription preference, blank spaces over invented content, local storage, distrust of cloud/model training, neutral tone, emojis reducing credibility	Classroom permission and consent, general customization, broad benefits, language-related barriers		Participants were open to AI support only when it remained accurate, non-inventive, privacy-conscious, and sufficiently restrained in what it generated.

Fig 1*: Thematic Comparison Table*

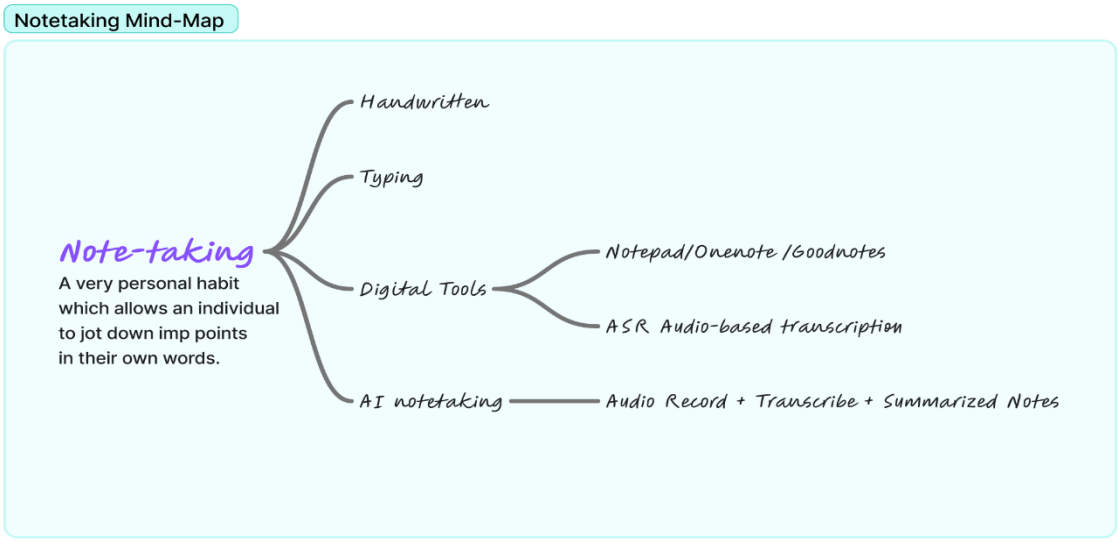


Fig 2: *Note-taking Modalities*

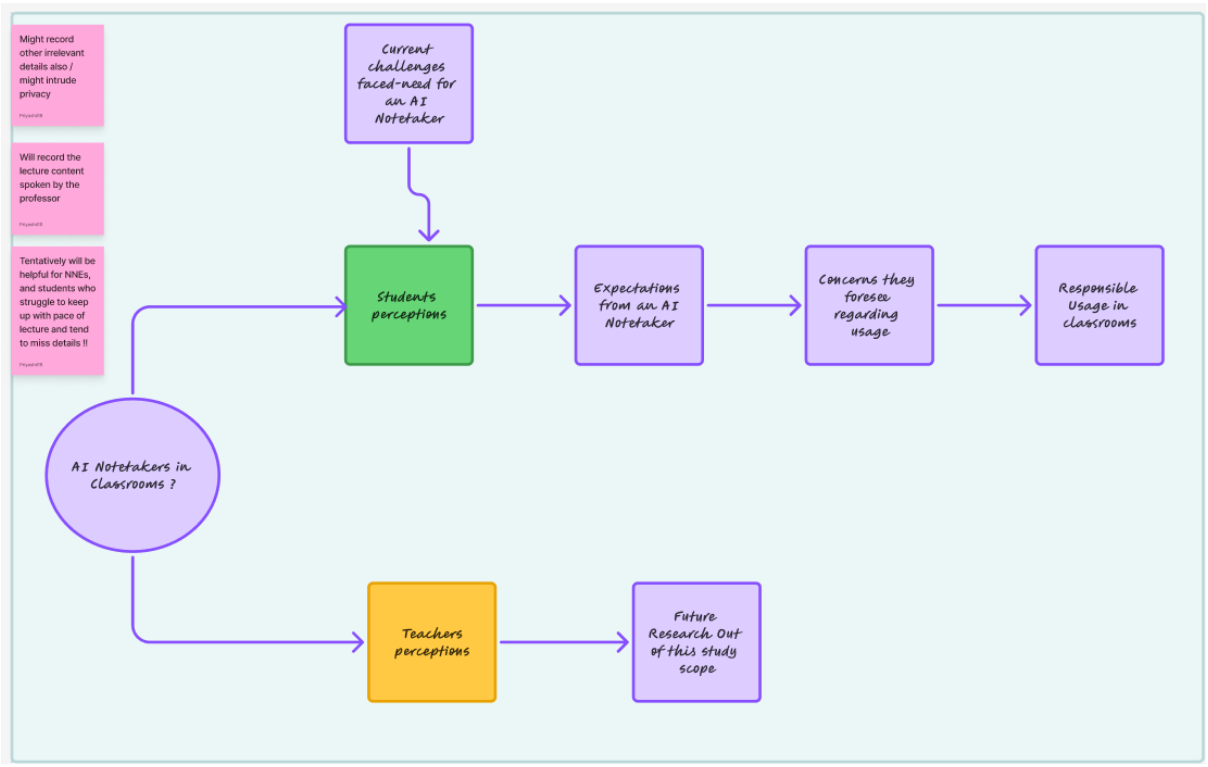


Fig 3: *Initial Framing of Study*

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Fig 4*: Interview Analysis on FigJam*

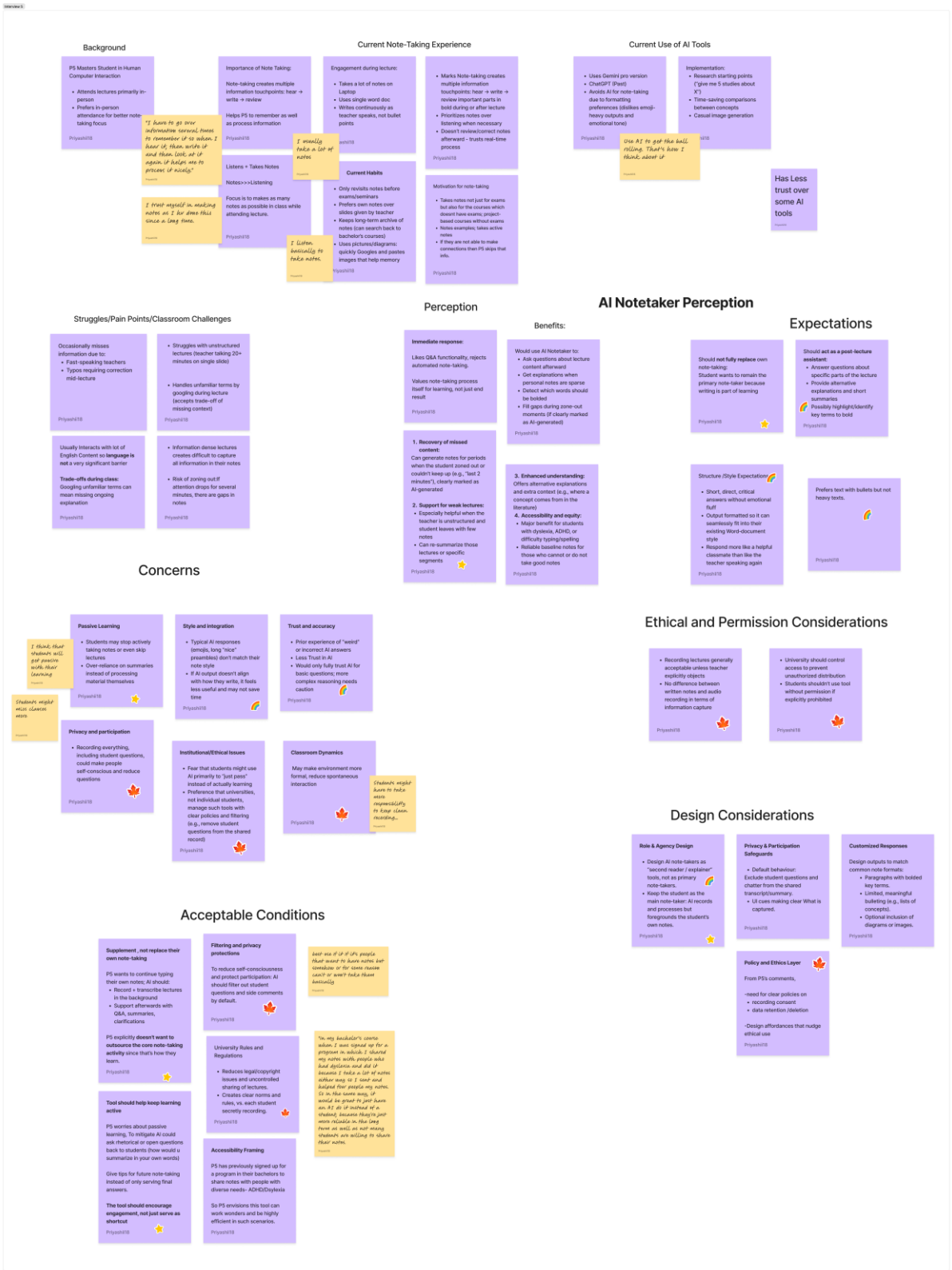


Fig 5: Interview Snippet from FigJam

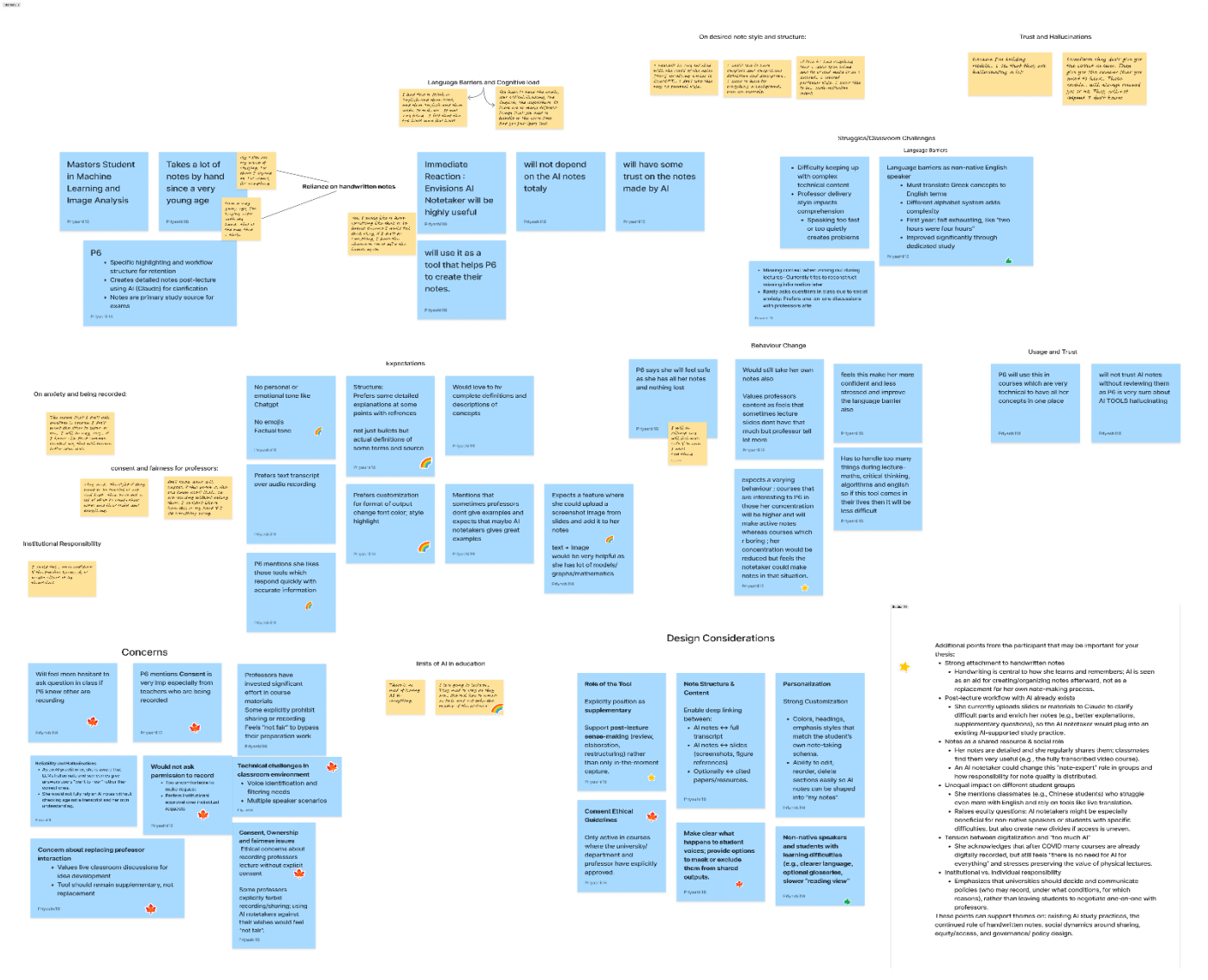


Fig 6 : Interview Snippet



Fig 7: Interview Analysis

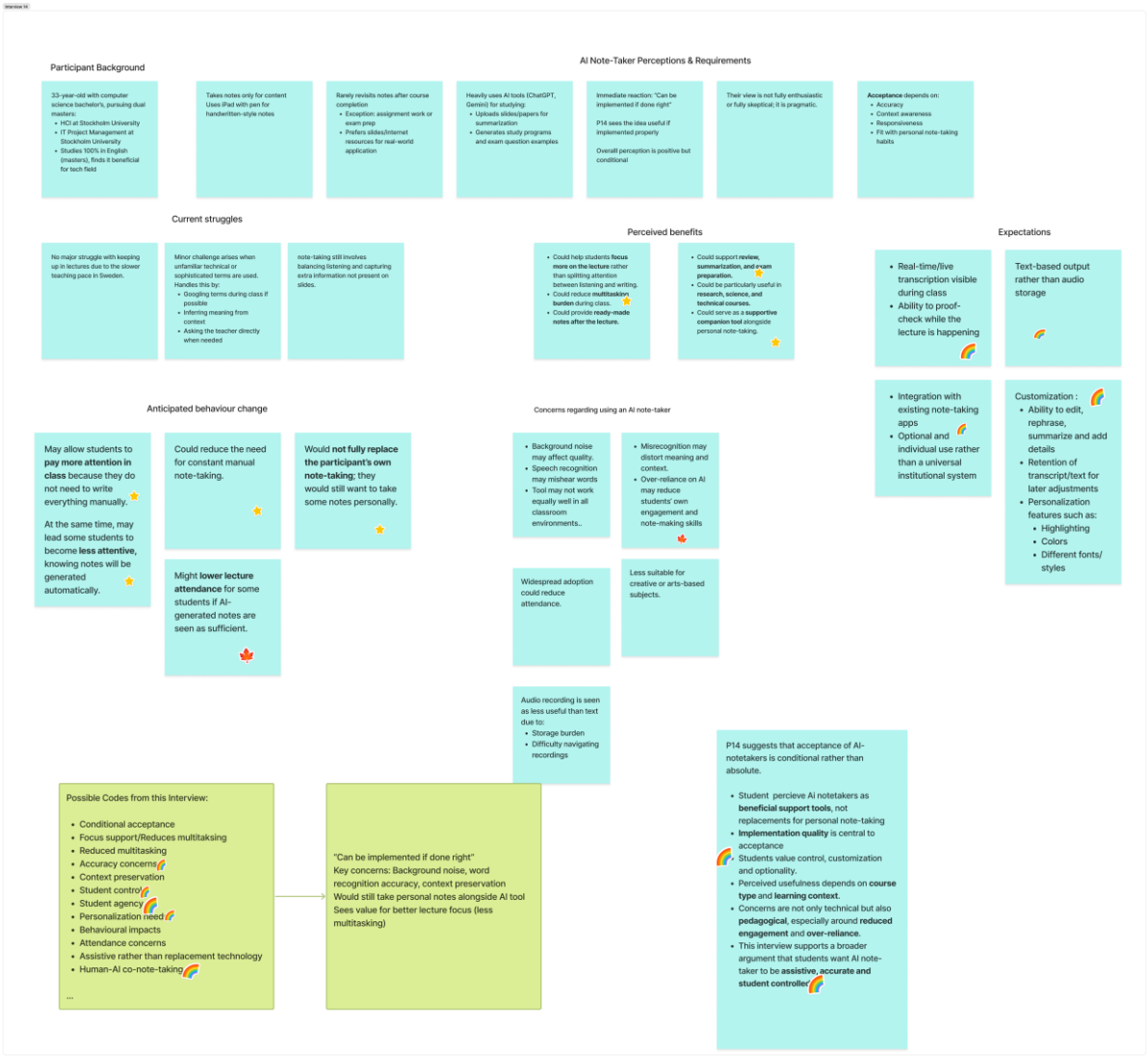
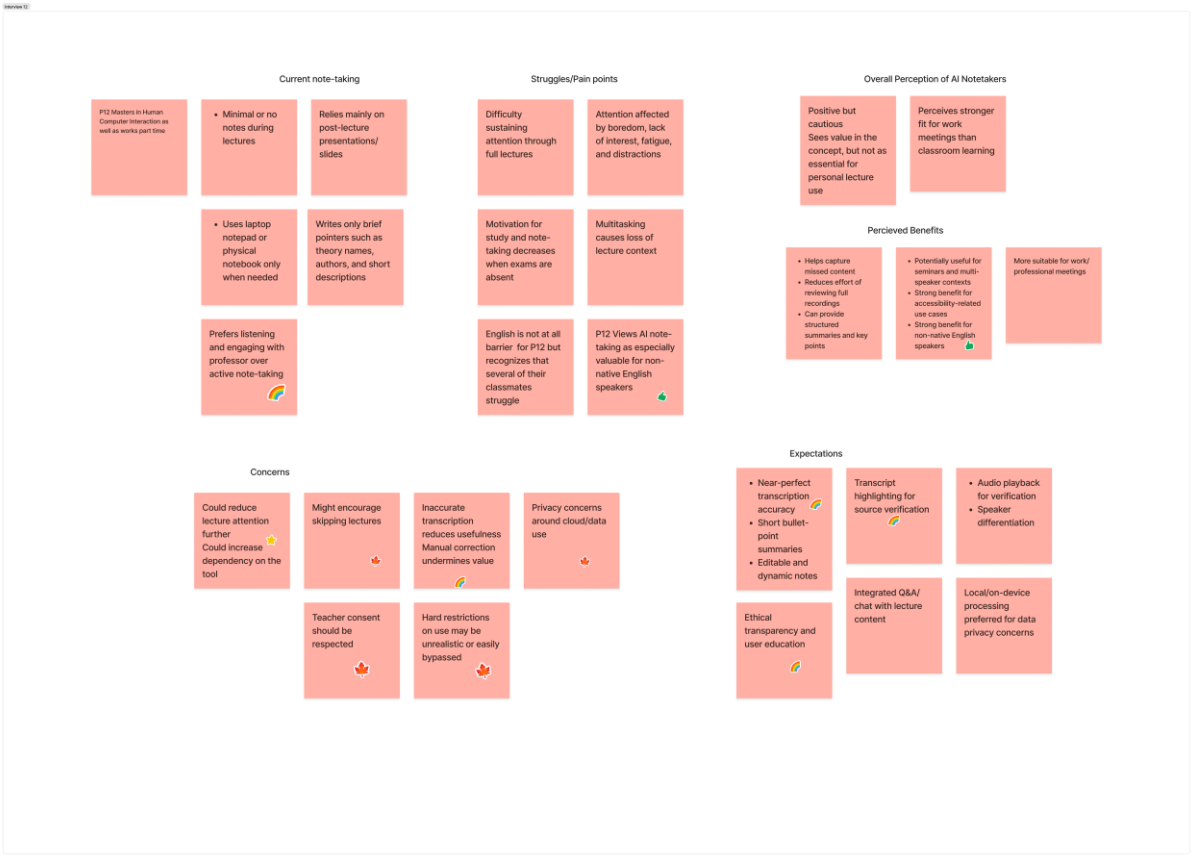


Fig 8: Interview Analysis


Interview 12
Current note-taking
P12 Masters in Human Computer Interaction as well as works part time
Minimal or no notes during lectures
Relies mainly on post-lecture presentations/ slides
Uses laptop notepad or physical notebook only when needed
Writes only brief pointers such as theory names, authors, and short descriptions
Prefers listening and engaging with professor over active note-taking
Struggles/Pain points
Difficulty sustaining attention through full lectures
Attention affected by boredom, lack of interest, fatigue, and distractions
Motivation for study and note-taking decreases when exams are absent
Multitasking causes loss of lecture context
English is not at all barrier for P12 but recognizes that several of their classmates struggle
P12 Views AI note-taking as especially valuable for non-native English speakers
Overall Perception of AI Notetakers
Positive but cautious
Sees value in the concept, but not as essential for personal lecture use
Perceives stronger fit for work meetings than classroom learning
Percieved Benefits
Helps capture missed content
Reduces effort of reviewing full recordings
Can provide structured summaries and key points
Potentially useful for seminars and multi-speaker contexts
Strong benefit for accessibility-related use cases
Strong benefit for non-native English speakers
More suitable for work/ professional meetings
Concerns
Could reduce lecture attention further
Could increase dependency on the tool
Might encourage skipping lectures
Inaccurate transcription reduces usefulness
Manual correction undermines value
Privacy concerns around cloud/data use
Teacher consent should be respected
Hard restrictions on use may be unrealistic or easily bypassed
Expectations
Near-perfect transcription accuracy
Short bullet-point summaries
Editable and dynamic notes
Transcript highlighting for source verification
Audio playback for verification
Speaker differentiation
Integrated Q&A/ chat with lecture content
Local/on-device processing preferred for data privacy concerns
Ethical transparency and user education

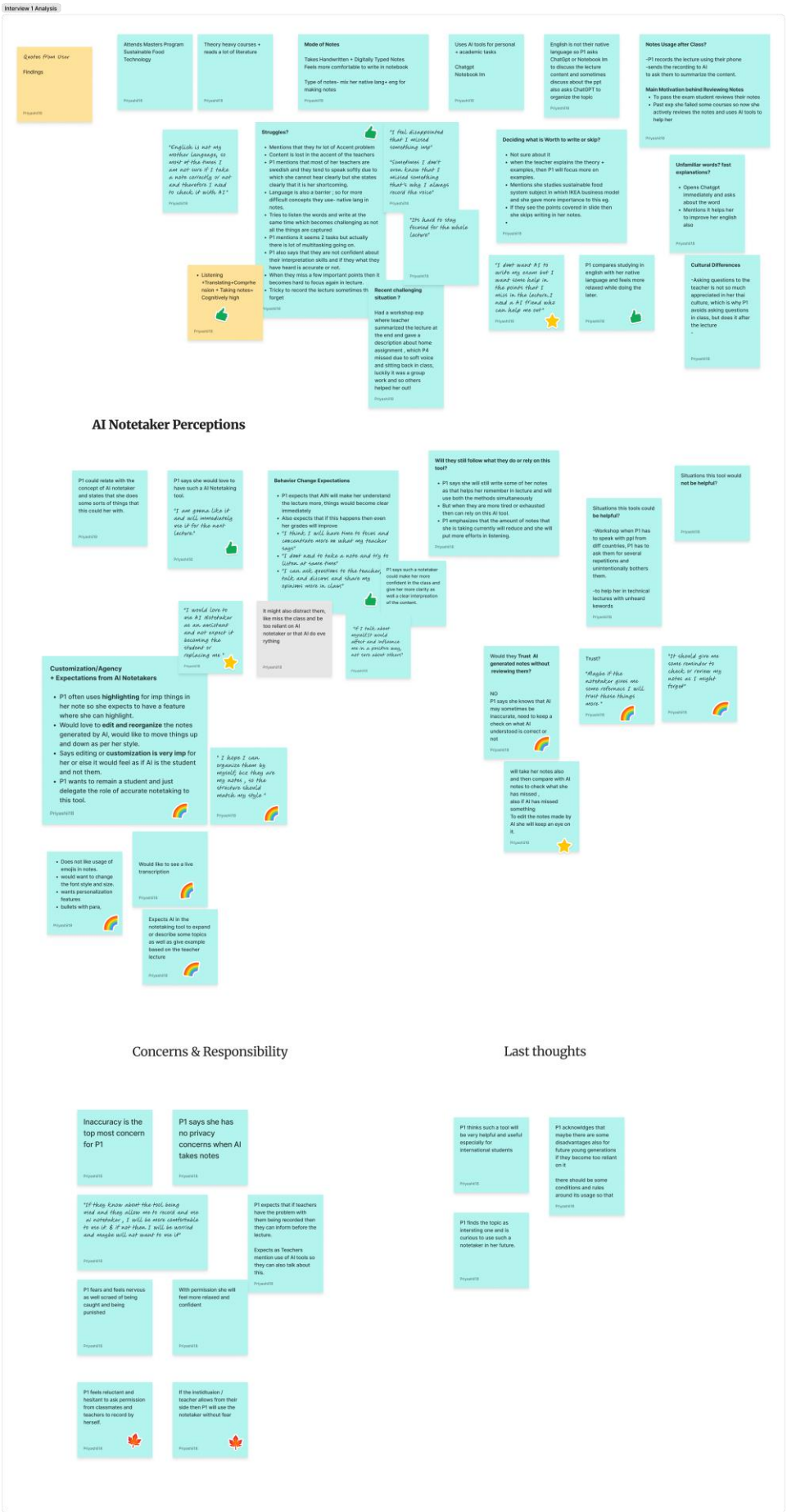


Fig 10: Analysis

MEMOS

- P5 is an active note-taker; who prefers note taking over listening.
- Mentions that he has to engage with information several times- listens, writes it down and then looks again at their notes while making it during the lecture which helps in remembering
- Uses AI for starting some ideation or to get the ball rolling but values critical thinking
- Though there are some struggles that they face like missing info; fast paced lectures but still prefers to write his own notes
- Expects to just ask the AI Notetaker some missed points or other scnarios
- doesnt want AI notetaker to replace his note taking
- Not a fan of passive learning
- P5 thinks about connections while taking notes from the lecture; skips content which does not make complete sense
- Asks questions to teacher and is not hesitant if the AI notetaker of classmates records them
- P4 and P5 both mentioned that they dont purposfully see other ppl's notes but if they do they do it for the structure not for copying the content (bullets with less text)
- P5 sees more drawbacks if this tool comes to class as they fear there are lots of ethical concerns which need to be addressed.
- P5 feels this tool could be of help if they zone out or not feeling well
- P5 understands that if AI notetaker comes in the classroom then students will also have to take more responsibility.
- At the end of the interaction P5 mentions that if AI notetaker are used in a controlled environment with approval from university and rules and regulations then it could be useful also for students who really need them.

Fig 11: Analytical Memos

MEMOS

- P3 initial reaction to AI Notetaker was regarding it being ethical to use in classroom and record without teacher's permission due to past experience.
- P3 was not sure if AI Notetaker could improve their focus more or reduce their concentration as writing notes helps them to remember and be concentrated in class- Ambivalent about it
- P3 will reduce their own note-taking mentions that they be more reliant on AI and would feel lethargic to do double work so will not only essential points which she thinks would be missed by AI ; like diagrams/graphs/charts/formulas.
- They prefer to personalize notes+would be unhappy if they dont have control over their notes. Agency matters!
- Emphasize about adding funny comments as part of her notes which makes their learning more engaging so would expect to do something like that with AI Notetaker also. - "would miss adding funny comments if AI notes are static"
- Expects a logical division of texts with clear visibility of headings+bullet points+ not text heavy as it is easier to read.
- No specific preference for font and styles.; highlighting is not very imp + uses her own abbreviations
- accuracy>>> structure + summarization
- P3 is skeptical about having trust on AI tools in general and which is the main reason they will not blindly trust notes made by AI but instead review them.
- P3 ellaborates that the lectures in which teachers explain or engage in an interactive and interesting story telling manner in those lectures she really wishes to be in the moment ; she expects this AI notetaker to act actively and take care of the note taking part while she can continue to focus; be fully present in the moment and enjoy her studies in classroom by listening attentively.
- Some teachers tend to be repeative in their class so she wishes that AI notetaker in that situation she hopes/wishes it could give her short and precise notes with no repetition.
- P3 considers themselves as shy student and mentions that if their classmates are using this kind a notetakers then that will make her more self concious ; more hesitant to ask/speak; will therefore avoid asking question to teacher.
- Ideally wants to see real time transcription but understands that in realistic scenarios it will be useful to hv it after class.
- Would prefer to use AI notetaker only if have consent from both teachers and students

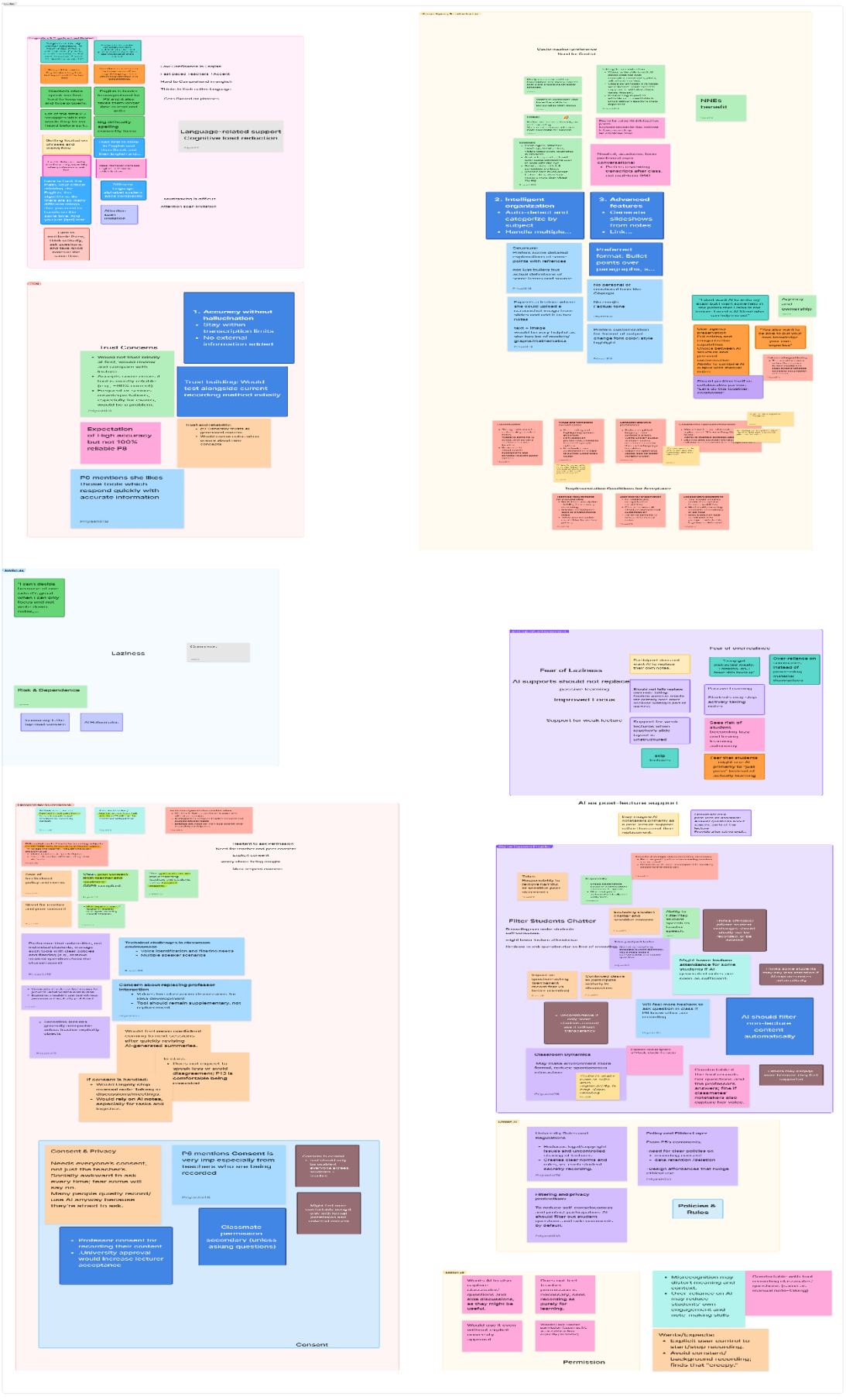


Fig 12: Codes from Data

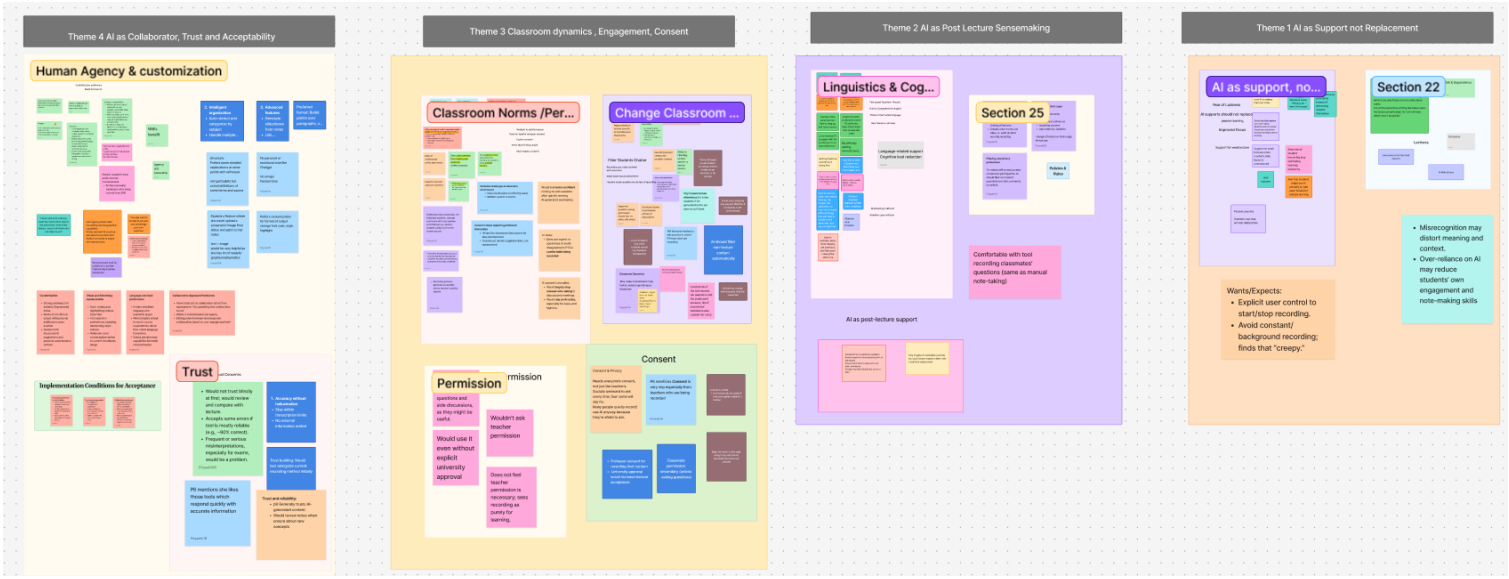


Fig 13: Four Overarching Themes Island Emerged

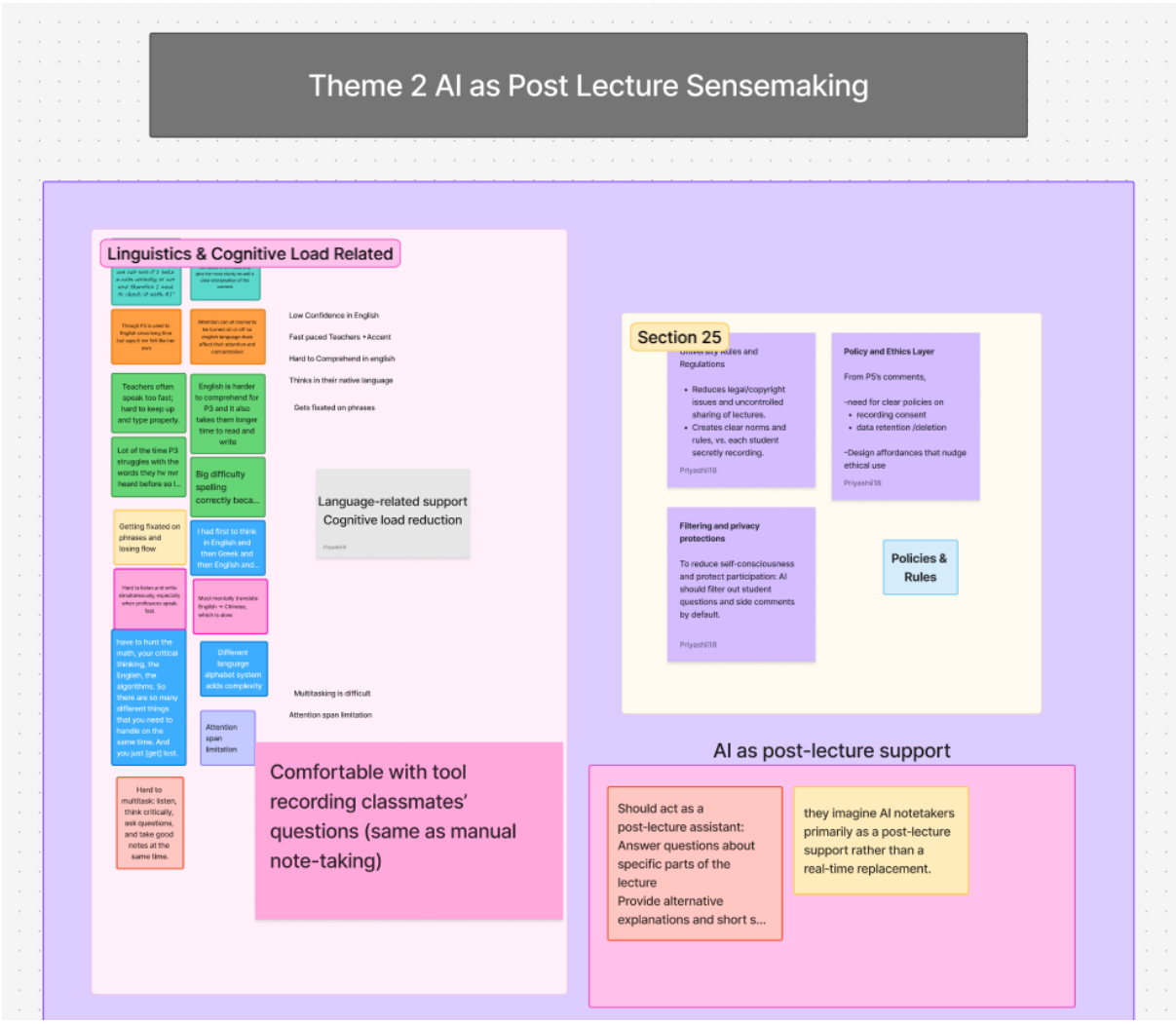


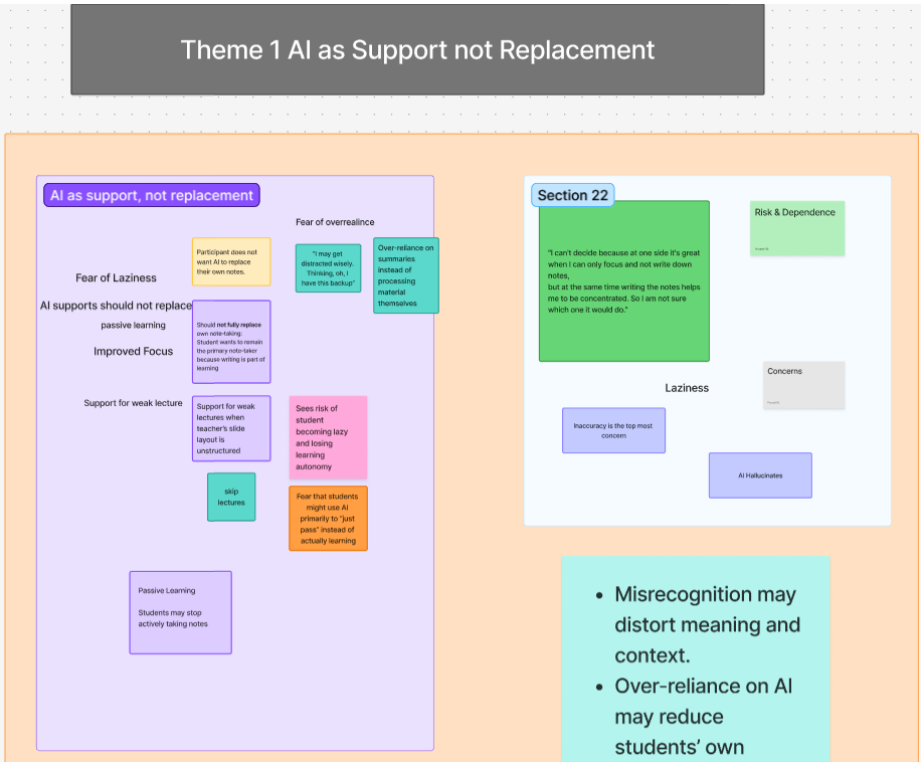
Fig 14 : Theme 2



Fig 15: Theme 3



Fig 16 : Theme 4



Fign 17: Theme 1

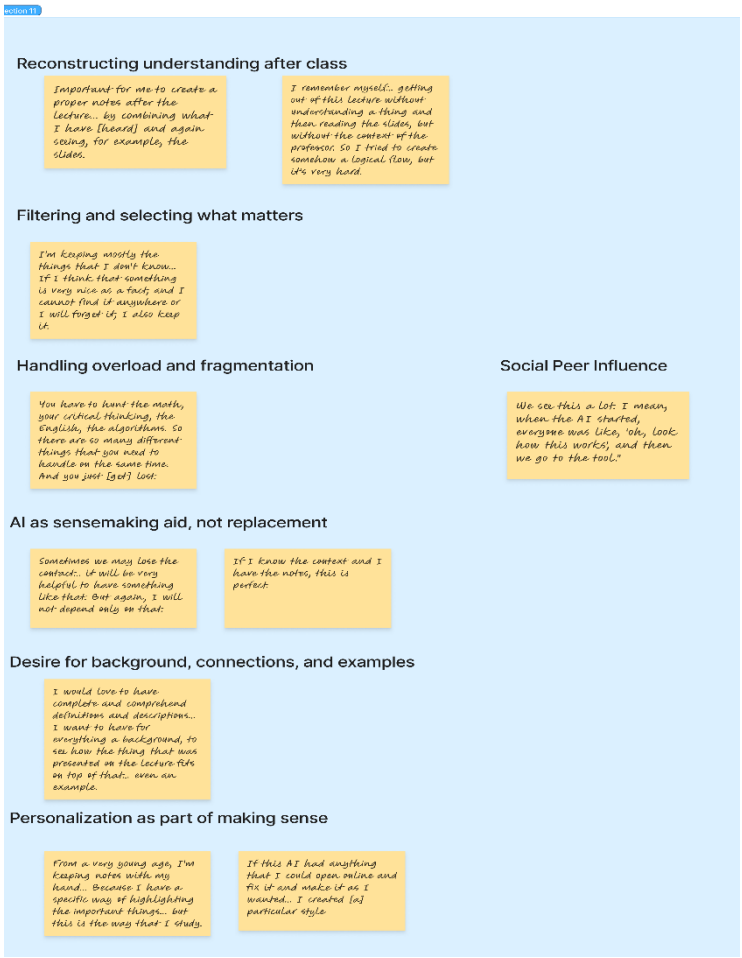


Fig 18: Participant Quotes

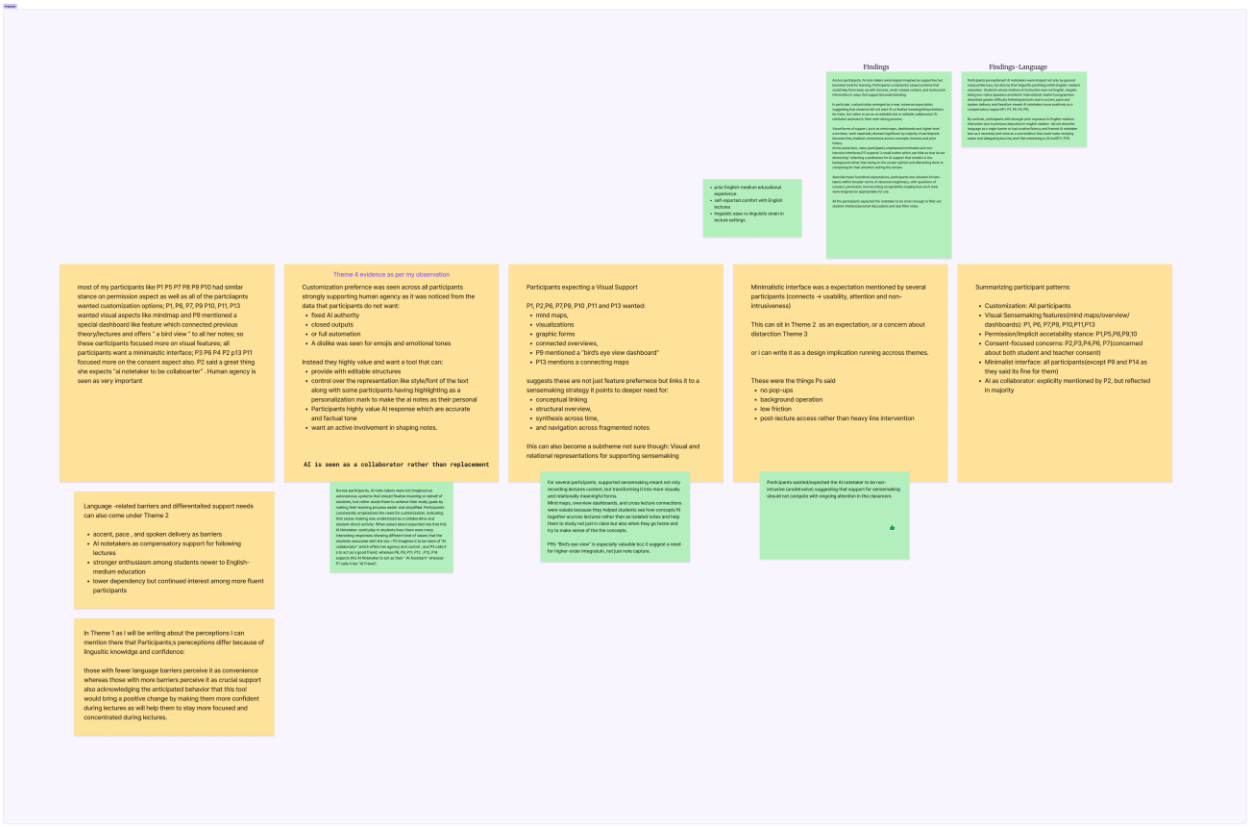


Fig 19: Results and Brainstorming

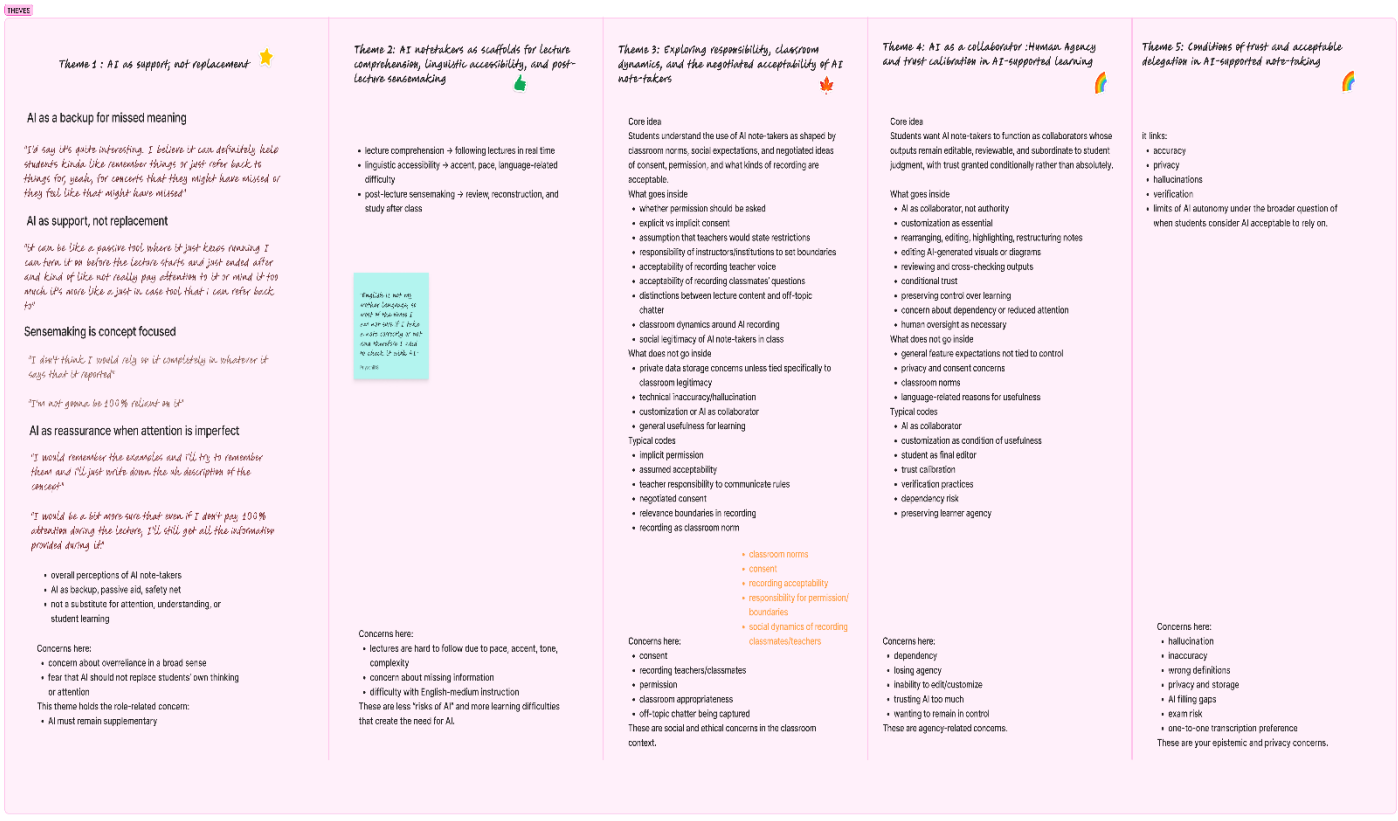


Fig 20: Initial themes formed

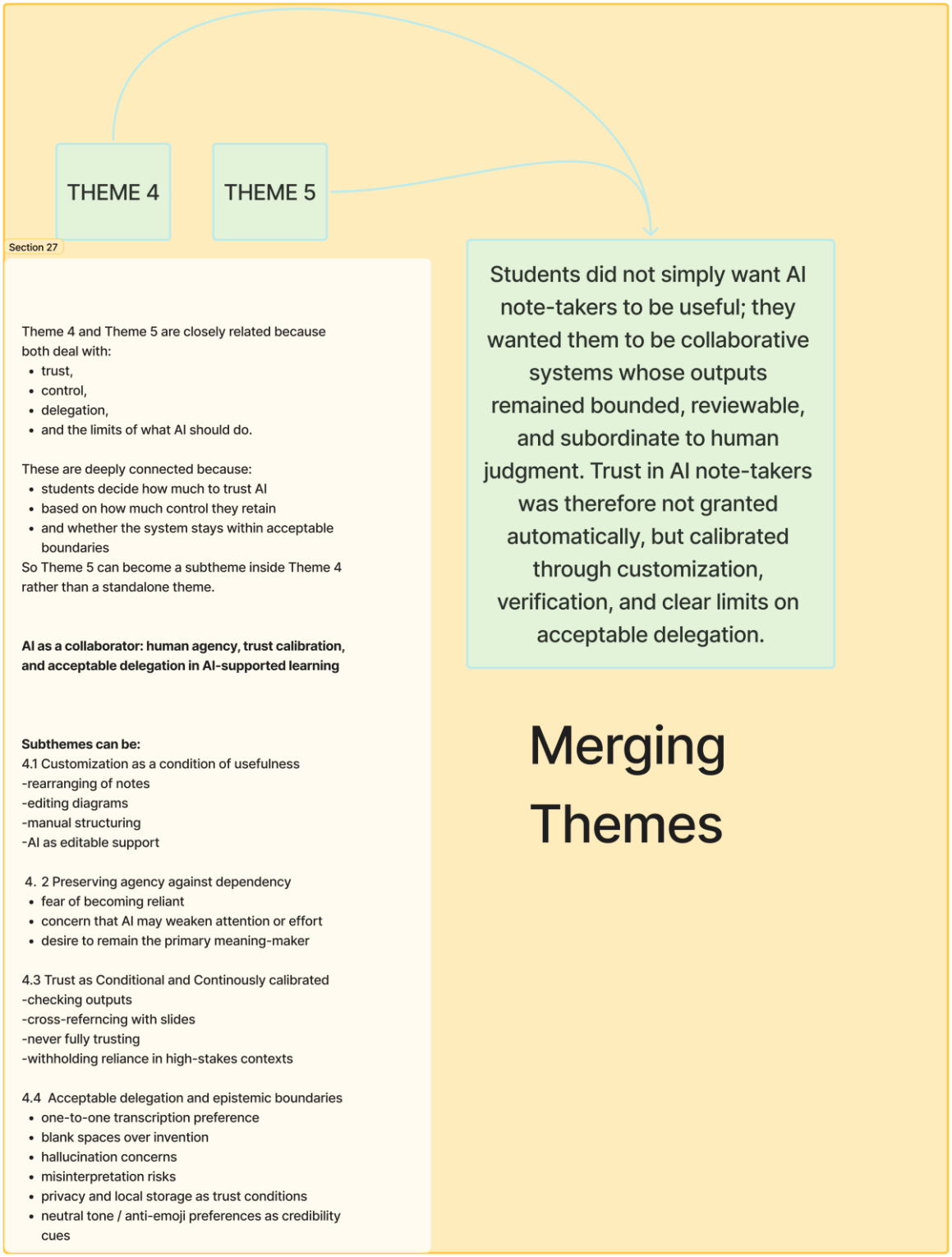


Fig 21: Merging Themes 4 and 5

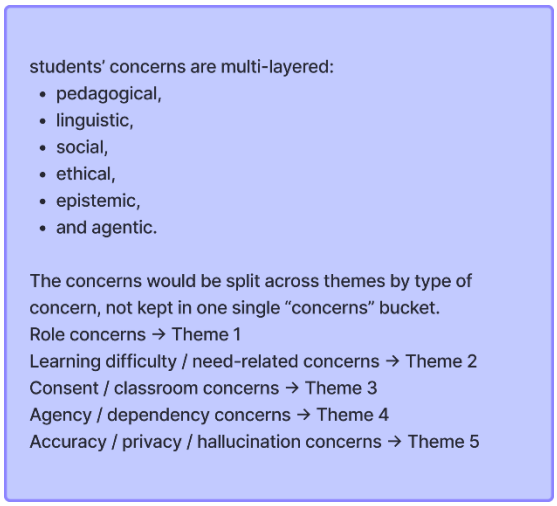


Fig 22: Concerns

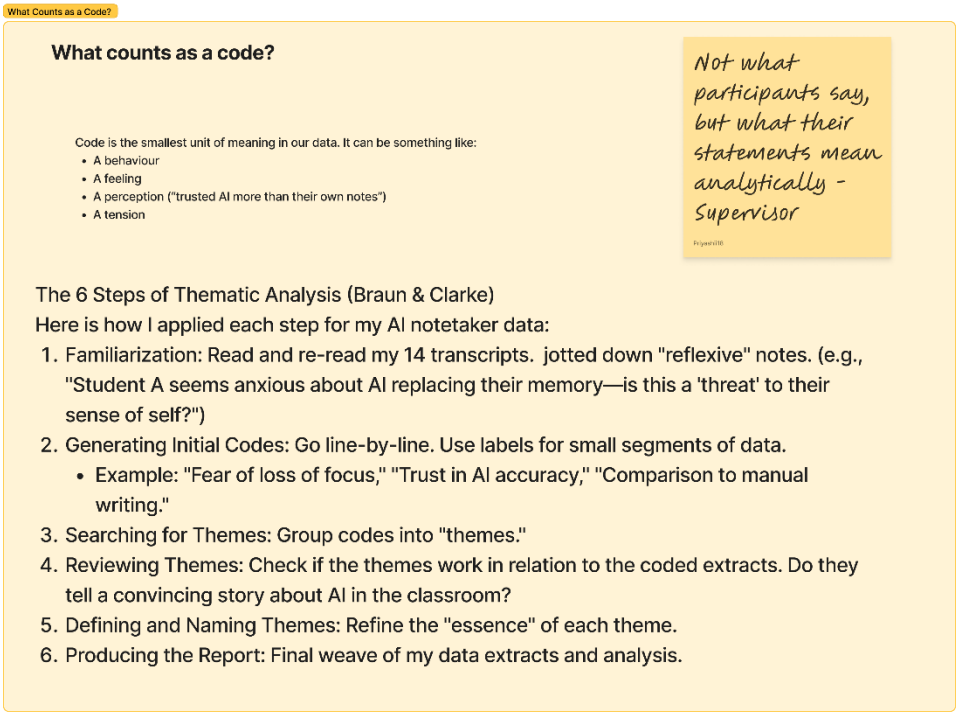


Fig 23: Thematic Analysis